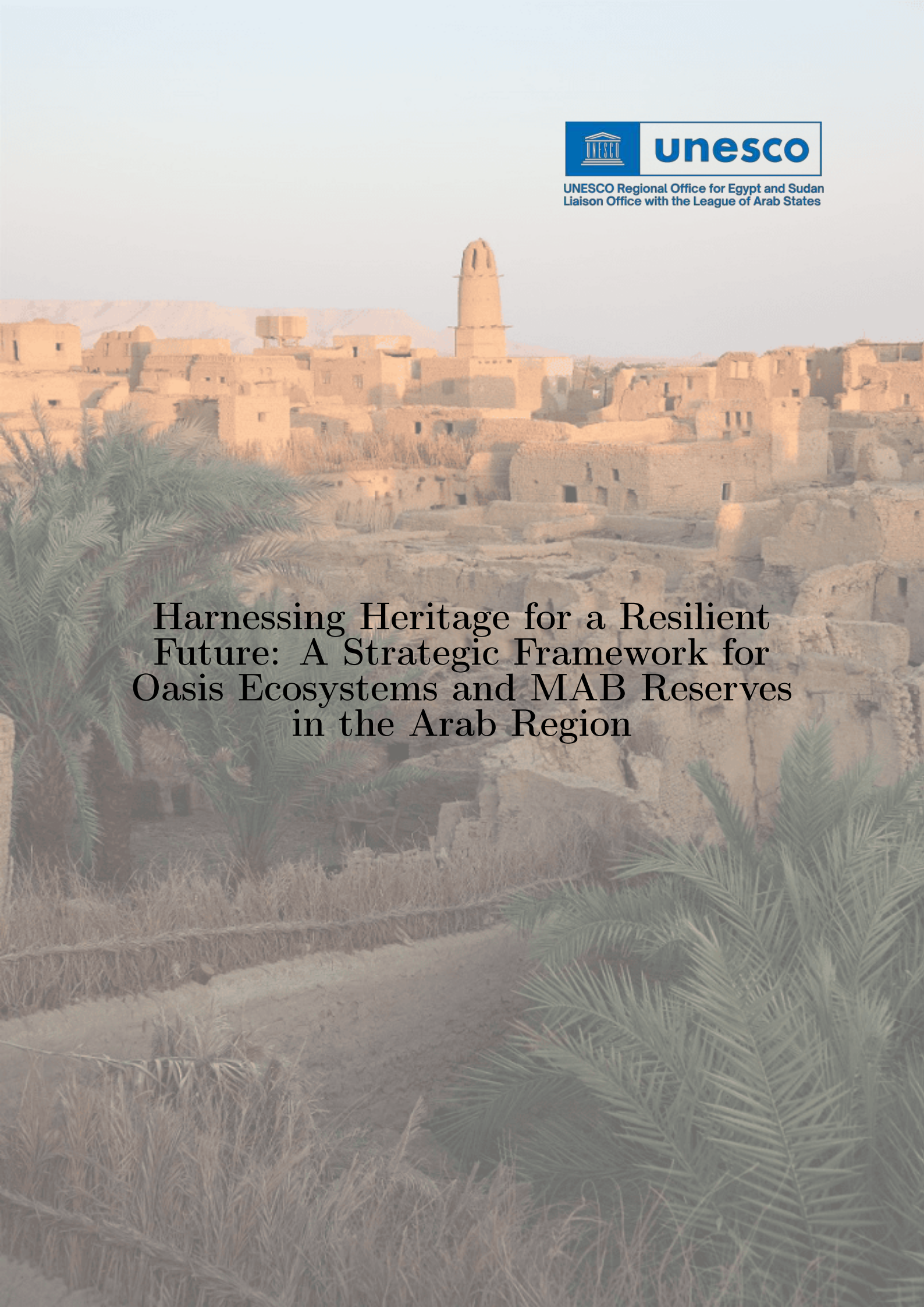

UNESCO Regional Office for Egypt and Sudan
Liaison Office with the League of Arab States

Harnessing Heritage for a Resilient Future: A Strategic Framework for Oasis Ecosystems and MAB Reserves in the Arab Region

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Executive Summary

Oases of the Arab region represent complex socio-ecological systems that have sustained human settlements and biodiversity in hyper-arid environments for millennia. These systems now face an existential crisis driven by groundwater depletion rates exceeding recharge by up to 1.1 billion cubic meters annually in some areas, climate change impacts reducing natural precipitation by nearly 30% in key regions [1], and socio-economic transformations that disrupt traditional management practices. This report presents a comprehensive strategic framework for oasis conservation and sustainable development, grounded in a paradigm shift from viewing oases as nature-based systems to recognizing them as heritage-based solutions where cultural practices, traditional knowledge, and social structures are integral to ecological resilience [2].

The framework establishes that the sustainability of oases depends on maintaining their biocultural diversity—the intricate co-evolution between biological systems and human cultural practices. Analysis reveals that linguistic diversity in oasis communities directly correlates with agrobiodiversity, with local languages containing precise terminology for traditional cultivation techniques and water management that have maintained these systems for centuries. The degradation of oases is not merely an environmental issue but a socio-cultural one, where the loss of traditional knowledge systems is as critical as aquifer depletion.

A systematic diagnosis of five major oasis systems across the Arab region reveals recurring patterns of failure: technological interventions without community governance lead to short-term gains followed by system collapse; top-down conservation approaches that exclude local knowledge result in partial restoration and community alienation; and unsustainable subsidies create perverse incentives that drive resource over-exploitation. Conversely, successful cases demonstrate that integrating traditional water management systems with modern governance structures, establishing strong legal frameworks with enforcement mechanisms, and creating economic incentives for traditional practices can enhance system resilience.

The strategic framework is built upon four integrated pillars:

1. An integrated and context-specific approach to water management that combines modern engineering with traditional systems
2. A foundational inventory and typology of oases to enable targeted planning
3. The centrality of indigenous knowledge and community empowerment as the primary drivers of sustainable management
4. Leveraging science and technology as enablers that complement rather than replace

Figure 1: Expert Group Meeting on Managing Groundwater and Dependent Ecosystems in Drylands (Oases), UNESCO Office in Cairo, June 2025

local wisdom

Priority actions include establishing a permanent Oasis Sub-Committee within the League of Arab States, developing a regional digital platform for knowledge management, creating economic models that value traditional practices through heritage tourism and certified products, and implementing a comprehensive monitoring framework with specific hydrological, ecological, and socio-economic indicators. By investing in the unique heritage of oases, the Arab region can preserve not only critical ecosystems but also a proven model of resilience for an increasingly arid world.

Foreword

The Arab Region's oases are not merely isolated pockets of green in vast deserts; they are living archives of human ingenuity and adaptation. For millennia, communities in these hyper-arid environments have thrived by mastering a delicate equilibrium between water, soil, and society. They developed sophisticated traditional ecological knowledge (TEK) systems—from the Aflaj water tunnels of Oman to the Khettaras of Morocco, from cooperative land management in Siwa to the precise water allocation mechanisms of the Draa Valley—that allowed civilization to flourish where survival itself seemed impossible.

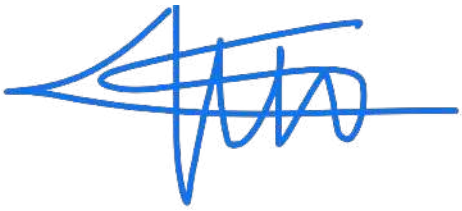
Today, however, we face a convergence of crises. Climate change, population growth, and groundwater depletion are straining these ancient systems to their breaking point. Extraction rates exceed recharge by billions of cubic meters annually. Springs that flowed for thousands of years have run dry. Traditional knowledge systems are eroding as rapidly as the aquifers themselves. Yet, a more subtle danger exists: the erosion of the cultural knowledge that once ensured resilience. When we treat oasis agriculture solely as an industrial output, or water management purely as hydraulic engineering, we lose the social and cultural fabric that makes these systems sustainable.

This strategic framework was conceived as a direct response to this challenge. It is grounded in a fundamental paradigm shift: recognizing oases not as nature-based systems requiring only technical interventions, but as heritage-based solutions where cultural practices, traditional knowledge, and social structures are integral to ecological function and resilience. We asked a critical question: Can we empower communities and policymakers to look back at their heritage not with nostalgia, but with a scientific eye, finding in it the keys to future sustainability?

We have adopted the integrated approach of sustainable development, but with a non-negotiable addition: the centrality of indigenous knowledge and community empowerment. We demand solutions that are scientifically rigorous yet deeply rooted in local contexts. We insist that modern technologies complement rather than replace traditional wisdom. We recognize that the sustainability of oases depends on maintaining their biocultural diversity—the intricate co-evolution between biological systems and human cultural practices.

The framework detailed in these pages is a testament to the collective expertise of regional and international specialists who participated in the Expert Group Meeting on Managing Groundwater and Dependent Ecosystems in Drylands. It documents our journey from diagnosis to prescription, from understanding systemic challenges to implementing concrete actions. It reflects UNESCO's commitment to nurturing these unique socio-ecological systems and the communities that have stewarded them for generations.

The time for action is now. This report provides the roadmap. The solutions are within reach—if we have the wisdom to learn from the past and the courage to embrace a new paradigm for the future.



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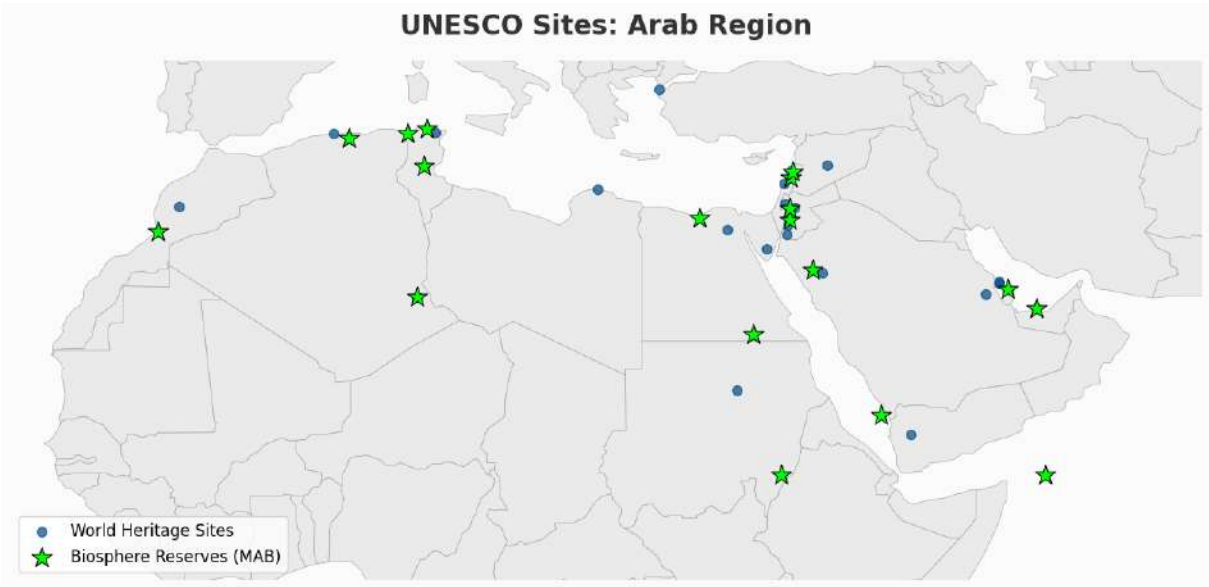


Figure 2: UNESCO Biosphere Reserves and World Heritage Sites in the Arab Region

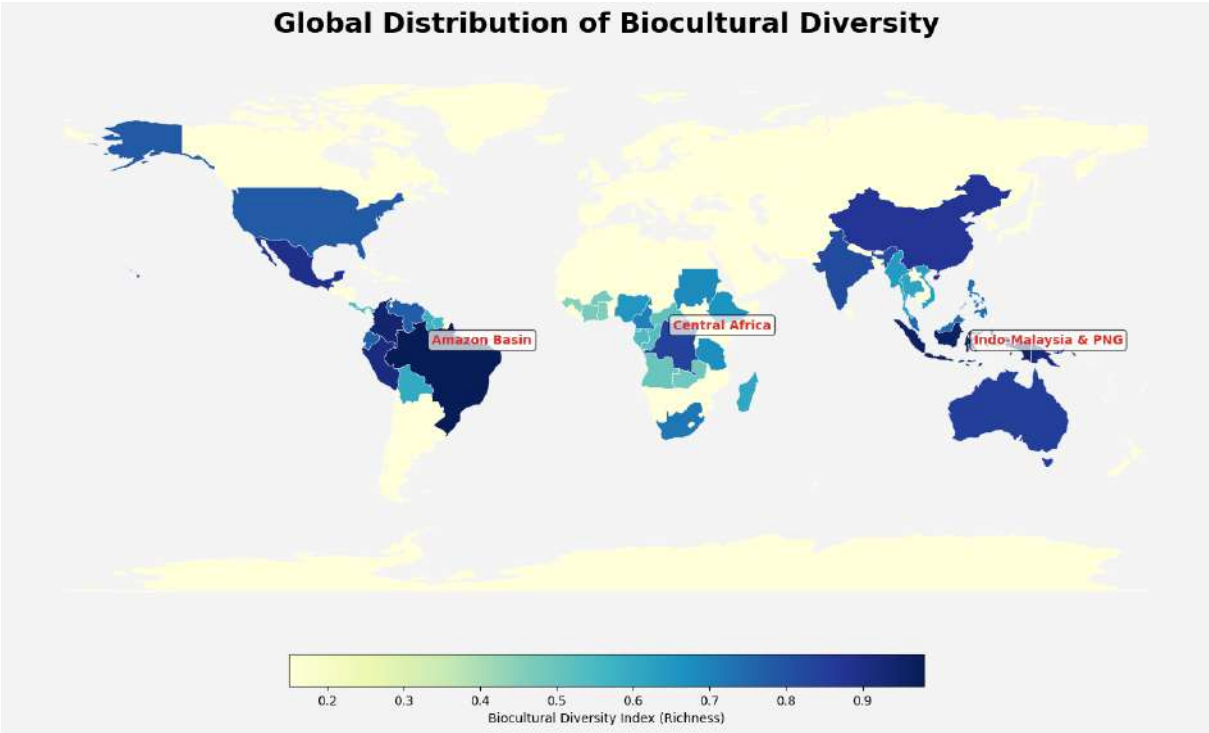


Figure 3: Global Distribution of Biocultural Diversity

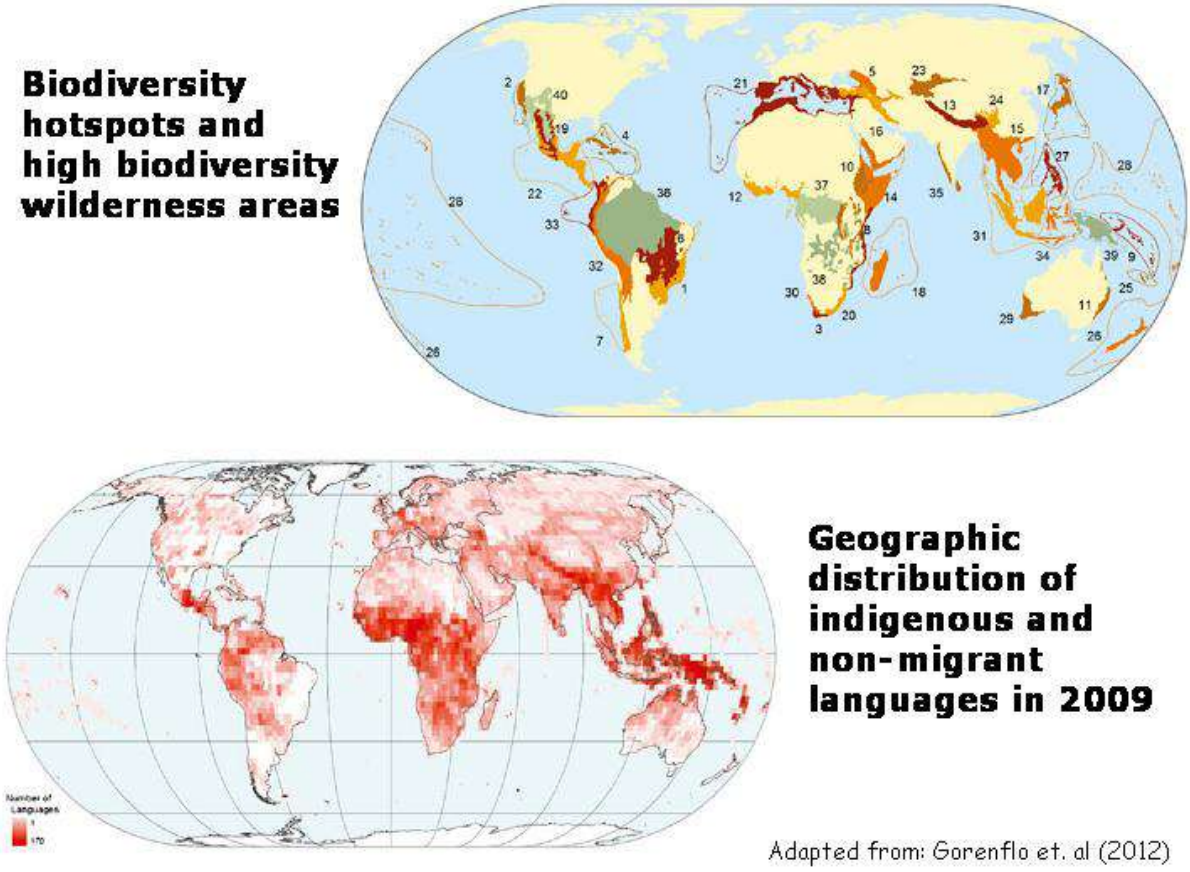


Figure 4: Linguistic Diversity and Biodiversity Correlation

Part I: The Foundation – The Oasis as a Critical Socio-Ecological System

Chapter 1

Introduction

The Arab Region encompasses some of the world's most arid environments, where water scarcity has shaped human settlement patterns for millennia. Within this challenging context, oases emerge as critical socio-ecological systems that have supported human civilizations, unique agricultural systems, and biodiversity hotspots. These systems are characterized by their dependence on groundwater resources, which in the Arab region are primarily sourced from non-renewable fossil aquifers such as the Nubian Sandstone Aquifer System (NSAS), or renewable but limited alluvial aquifers recharged by seasonal rainfall.

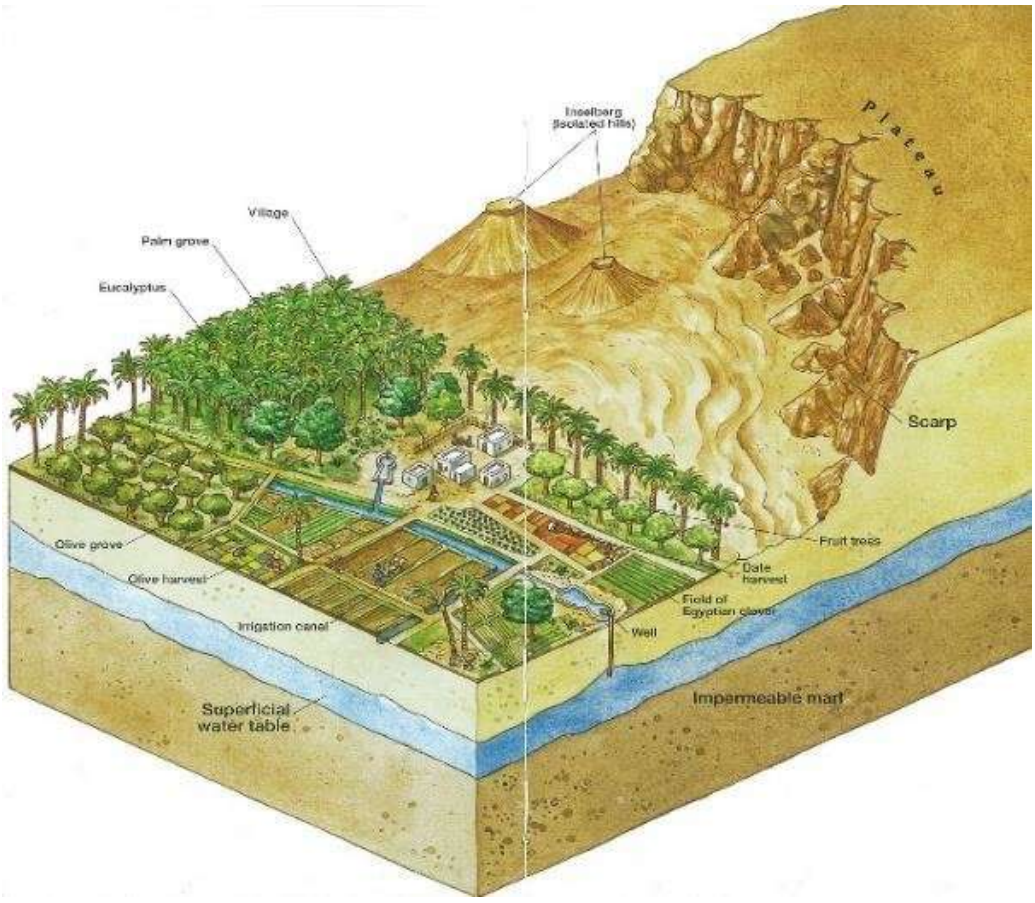


Figure 1.1: Oasis Ecosystems: Critical Components and Functions

Oases function as complex adaptive systems where hydrological processes, ecological dynamics, and human activities are intricately interconnected. The sustainability of these systems depends on maintaining equilibrium between water extraction and recharge, soil fertility management through organic matter recycling, and biodiversity conservation through traditional agricultural practices. However, this equilibrium has been increasingly disrupted by anthropogenic pressures, with groundwater extraction rates in many areas exceeding natural recharge by factors of 3-5 times, leading to declining water tables, increased salinization, and ecosystem degradation.

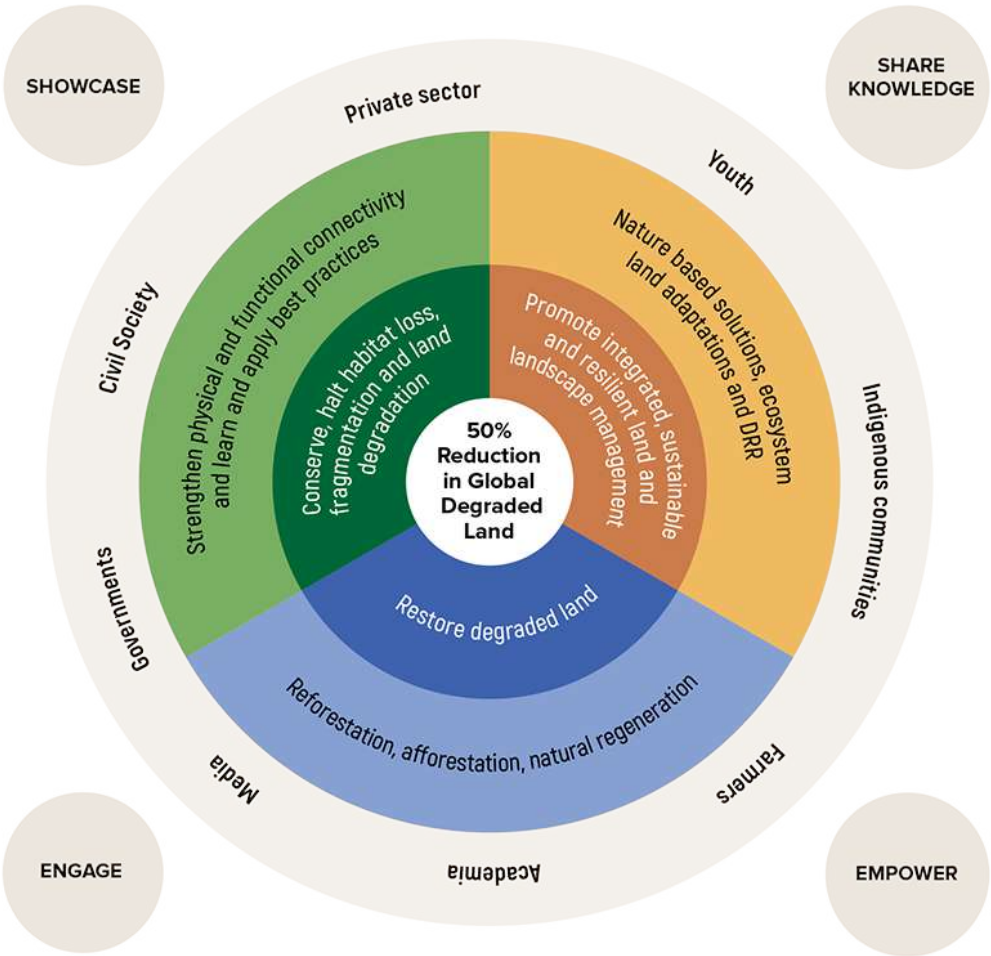


Figure 1.2: G20 Global Initiative on Ecosystem Restoration and Land Degradation Reduction

Climate change further exacerbates these challenges, with projections indicating temperature increases of 2-4°C in the region by 2050, accompanied by increased evaporation rates and more frequent extreme weather events [1]. These changes directly impact groundwater recharge patterns and agricultural productivity, creating additional stress on already vulnerable systems. The combination of water scarcity, climate change, and socio-economic transformations has created a crisis that threatens the very existence of these millennia-old systems.

This report provides a comprehensive strategic framework for the sustainable management of oasis ecosystems in the Arab region, with particular emphasis on their integration into

the UNESCO Man and the Biosphere (MAB) Programme. The framework is grounded in a paradigm shift that recognizes oases not merely as natural systems but as cultural landscapes shaped by human ingenuity and traditional knowledge. It presents a detailed analysis of current challenges, a comparative assessment of management approaches, and a concrete action plan for enhancing the resilience of these critical systems [3].

The report is structured to provide policymakers, program managers, and technical partners with a comprehensive understanding of oasis systems and practical guidance for their conservation and sustainable development. It synthesizes current scientific understanding with traditional knowledge, offering a pathway to secure the future of these unique socio-ecological systems.

Chapter 2

The Oasis Paradigm - A Heritage-Based Solution

The conventional approach to oasis conservation has focused primarily on their hydrological and ecological characteristics, treating them as natural systems to be managed through technical interventions. This perspective has proven inadequate, as evidenced by the continued degradation of oasis systems despite significant investments in infrastructure and technology. A more effective paradigm recognizes oases as heritage-based solutions—socio-ecological systems where cultural practices, traditional knowledge, and social structures are integral to ecological function and resilience.

2.1 Biocultural Diversity as the Foundation of Oasis Resilience

Biocultural diversity—the interdependent relationship between biological and cultural diversity—emerges as the critical measure of oasis health. In oasis systems, this relationship manifests in multiple dimensions:

Genetic-Linguistic Correlations: Research in the Siwa Oasis demonstrates a direct correlation between linguistic diversity and genetic diversity of date palms [4]. The local Amazigh language contains specific terminology for traditional cultivation techniques, water management practices, and genetic selection processes that have maintained and enhanced date palm diversity over centuries. This linguistic precision reflects generations of accumulated knowledge about micro-climatic variations, soil types, and water quality that would be difficult to capture through conventional scientific approaches alone.

Traditional Knowledge Systems: The sophisticated water management systems found in oases—such as the Omani Aflaj, Moroccan Khettaras, and Persian Qanats—represent complex socio-hydrological systems that integrate engineering, ecology, and social organization. These systems incorporate detailed understanding of groundwater hydrology, soil mechanics, and plant physiology, embedded within social structures that ensure equitable distribution and sustainable use. The Aflaj system, for instance, employs precise water allocation mechanisms based on time units measured by stellar observations, with different water qualities allocated for specific uses according to a hierarchy of needs.

Figure 2.1: Date Palm Agrobiodiversity and Traditional Knowledge Systems in Oasis Ecosystems

Cultural Practices and Ecosystem Services: Traditional agricultural practices in oases—such as three-tiered cultivation (date palms, fruit trees, and ground crops), crop rotation, and organic fertilization—enhance biodiversity, improve soil fertility, and optimize water use efficiency. These practices are not merely production techniques but cultural institutions that regulate human-environment interactions, maintain landscape heterogeneity, and provide multiple ecosystem services including food security, microclimate regulation, and biodiversity conservation.

2.2 The Hydrological-Cultural Nexus

The hydrological functioning of oases cannot be separated from their cultural dimensions. Traditional water management systems incorporate sophisticated understanding of:

Groundwater Dynamics: Traditional communities possess detailed knowledge of aquifer characteristics, including recharge zones, flow paths, and discharge points. This knowledge, accumulated through generations of observation and experience, often corresponds closely with scientific hydrogeological understanding but includes additional insights about temporal variations and ecosystem responses that are difficult to capture through conventional monitoring.

Water Quality Management: Traditional systems incorporate sophisticated water quality management practices, including the use of specific plant species as bio-indicators of water quality, the separation of water uses based on quality gradients, and the management of drainage to prevent salinization. In the Siwa Oasis, for instance, traditional practices included the management of saline lakes to extract salt for preserving olives, transforming a potential problem into a valuable resource [4].

Climate Resilience: Traditional architecture and settlement patterns in oases demonstrate sophisticated adaptation to climatic extremes, including passive cooling systems,

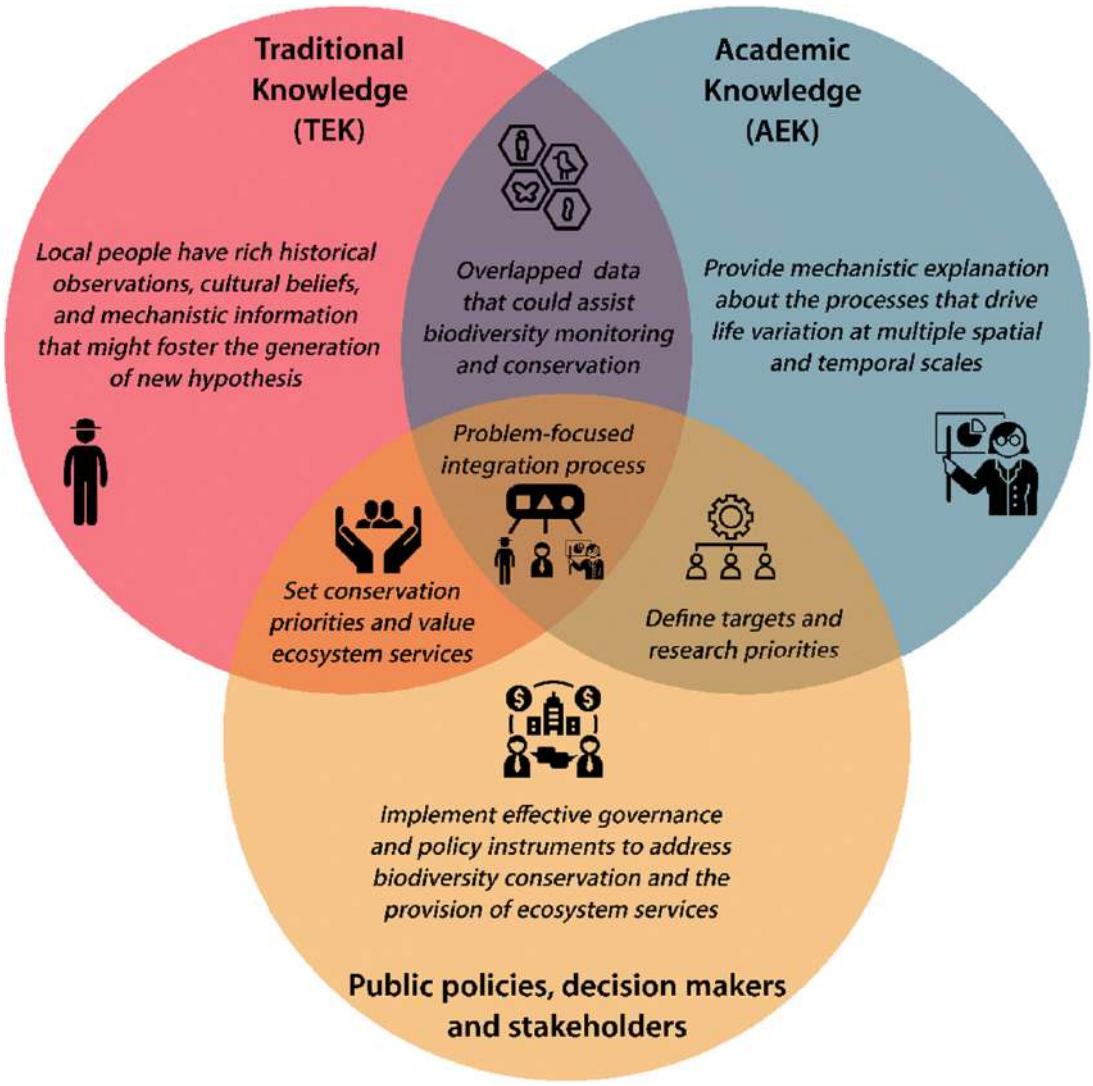


Figure 2.2: Integration of Traditional Ecological Knowledge into Academic Research

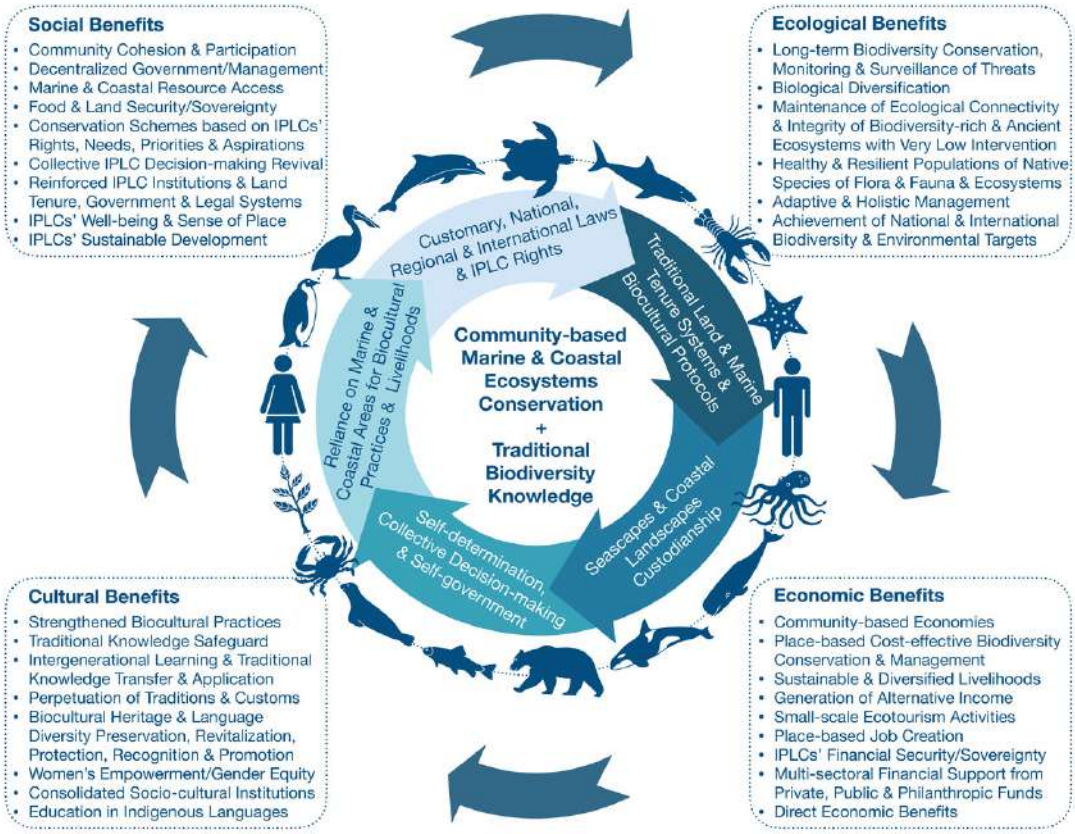


Figure 2.3: UNESCO Framework for Biodiversity Conservation Through Traditional Knowledge

windbreaks, and water harvesting techniques that enhance resilience to temperature extremes and water scarcity. These practices are not merely historical artifacts but living traditions that continue to evolve in response to changing conditions.

This heritage-based paradigm reorients conservation strategies from purely technical interventions to approaches that strengthen the social and cultural foundations of oasis systems. It recognizes that the sustainability of oases depends on maintaining the integrity of their biocultural diversity and the adaptive capacity of their associated knowledge systems.

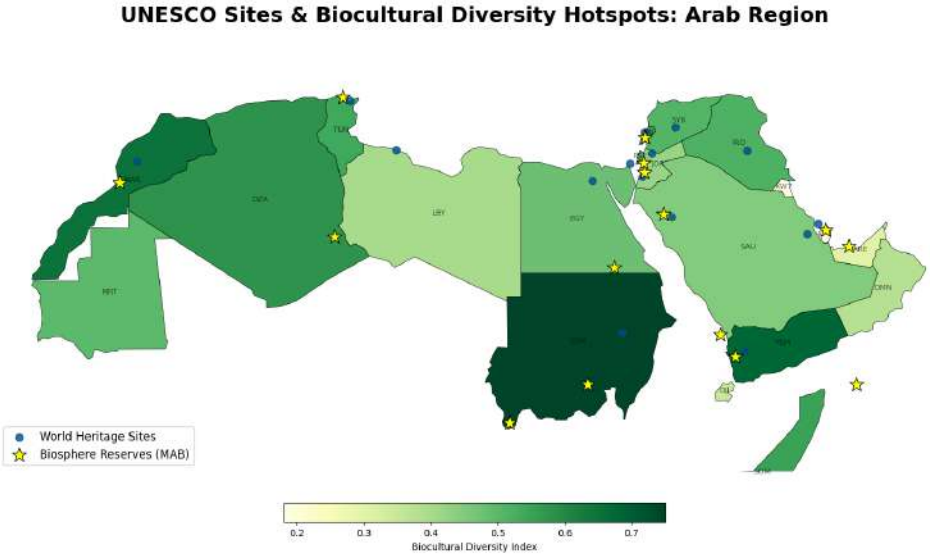


Figure 2.4: Map of Biocultural Diversity and UNESCO Sites in the Arab Region

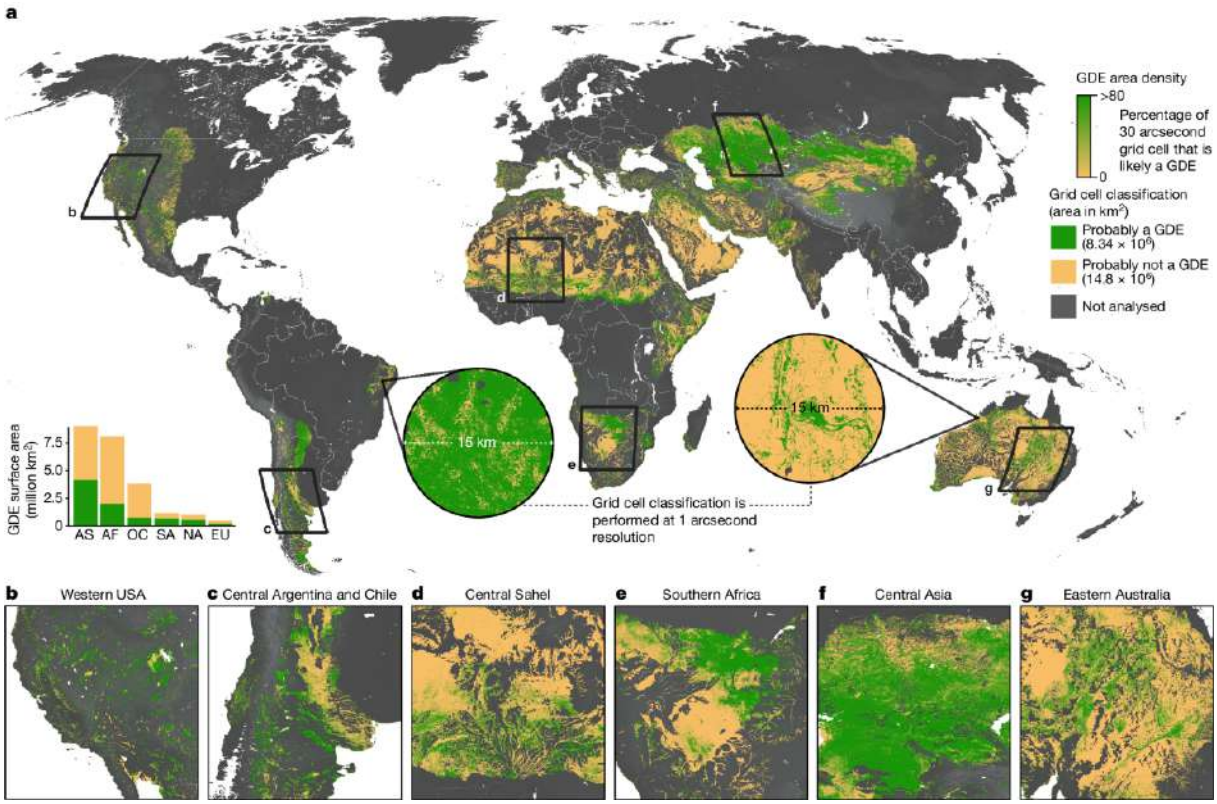


Figure 2.5: Groundwater Dependent Ecosystems in Drylands: Theory and Practice

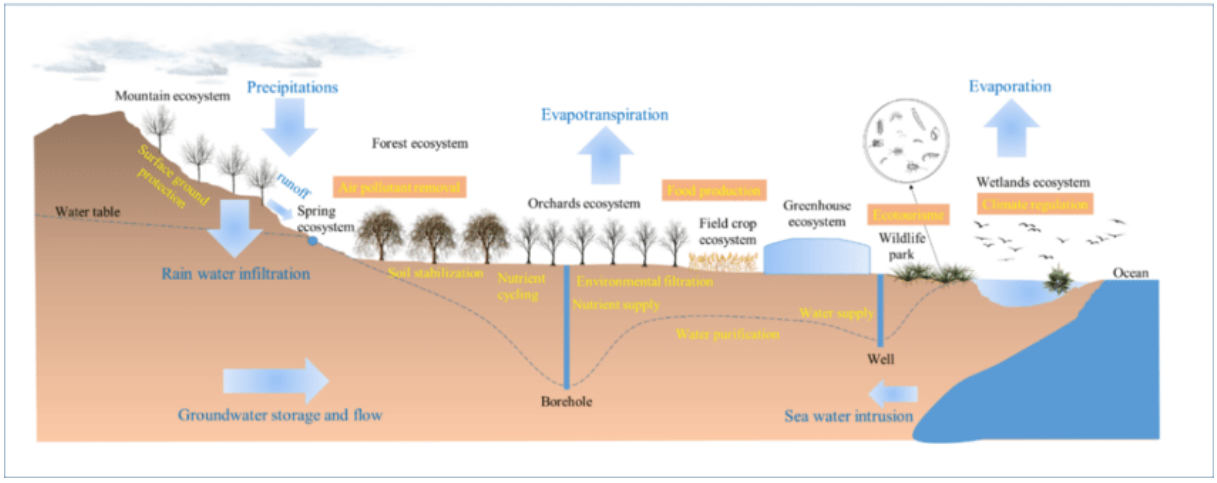


Figure 2.6: Challenges Faced by Groundwater-Dependent Ecosystems

Part II: The Diagnosis – Pressures, Weaknesses, and Resilience

Chapter 3

A Typology and Status of Arab Oases

A comprehensive understanding of oasis systems requires a systematic classification that captures their diversity in terms of hydrogeological characteristics, socio-economic functions, and developmental trajectories. Based on expert analysis and field research, oases in the Arab region can be classified according to the following typology:

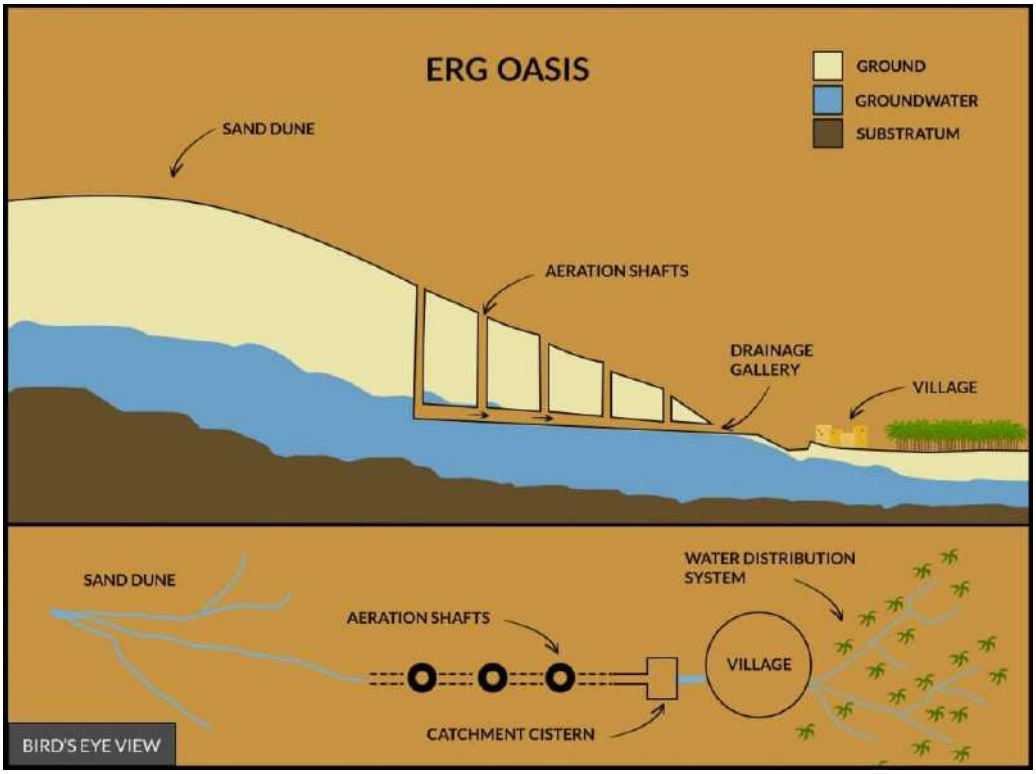


Figure 3.1: Traditional Oasis Water Management: Cultural Heritage Preservation

3.1 Hydrogeological Classification

Spring-Fed Oases: These systems depend on natural springs discharging from aquifers where groundwater pressure is sufficient to reach the surface without mechanical extraction. Examples include the historical springs of the Siwa Oasis in Egypt and the Azraq

Oasis in Jordan. These systems are particularly vulnerable to groundwater level declines, as even small reductions in pressure can lead to spring failure. The hydrological dynamics of spring-fed oases are characterized by stable baseflow conditions with limited capacity to respond to increased demand.

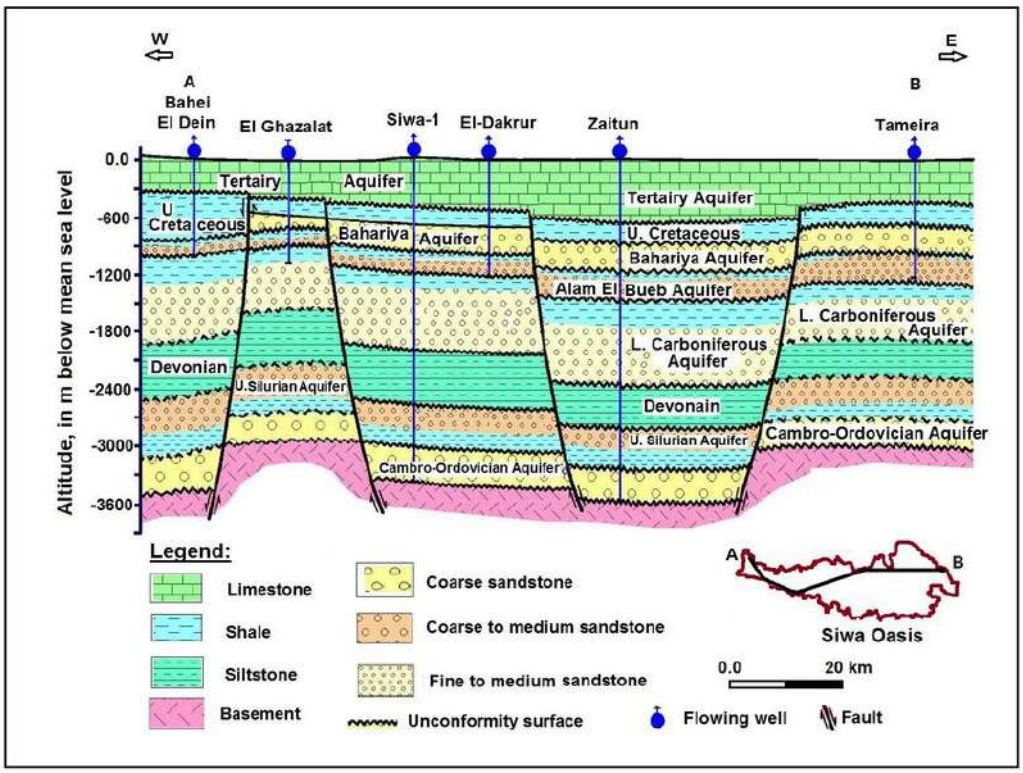


Figure 3.2: Hydrogeological Characteristics of Major Oasis Systems

Qanat/Khattara-Fed Oases: These systems utilize gravity-flow underground tunnels (qanats in Iran, khettaras in Morocco, aflaj in Oman) that tap into shallow aquifers and convey water to the surface by gentle gradients. These systems represent sophisticated engineering achievements that require minimal energy input and provide consistent water flow. The hydrological characteristics include relatively stable discharge rates, limited susceptibility to evaporation losses, and dependence on the maintenance of tunnel structures and water distribution networks.

Pumped Aquifer Oases: Modern oasis systems increasingly depend on pumped extraction from deep aquifers using diesel or electric pumps. These systems can access larger volumes of water from deeper aquifers but require significant energy inputs and infrastructure investment. The hydrological dynamics are characterized by declining water tables, increasing pumping lifts, and deteriorating water quality over time. Examples include the New Valley oases in Egypt, which depend on the non-renewable Nubian Sandstone Aquifer System [5].

Mixed-Source Oases: Many contemporary oasis systems utilize multiple water sources, combining traditional methods with modern pumping, surface water diversions, and increasingly, treated wastewater. These systems exhibit complex hydrological interactions between different water sources and require integrated management approaches. The Draa Valley in Morocco represents such a system, combining surface water from the El Mansour Eddahbi dam with groundwater extraction during drought periods.

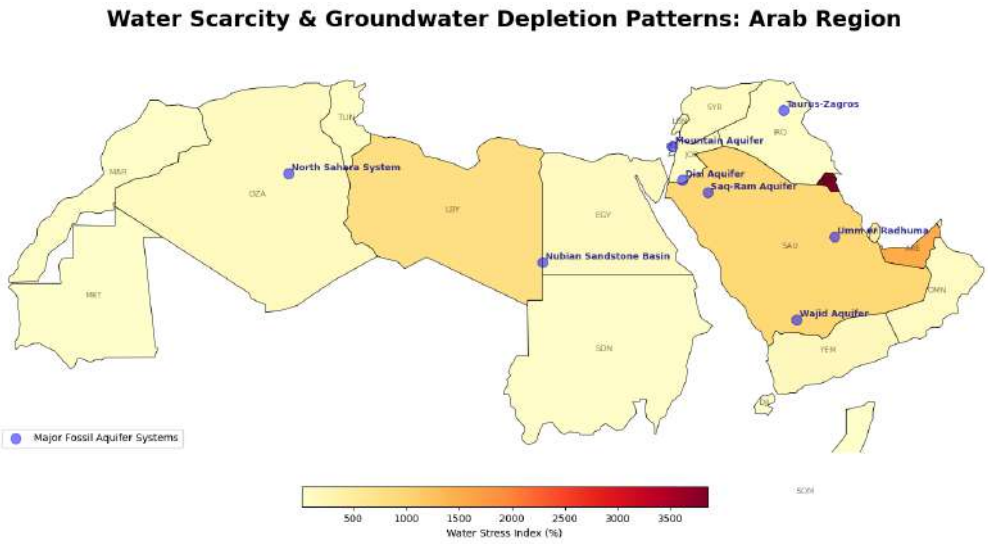


Figure 3.3: Water Scarcity and Groundwater Depletion Patterns Across the Arab Region

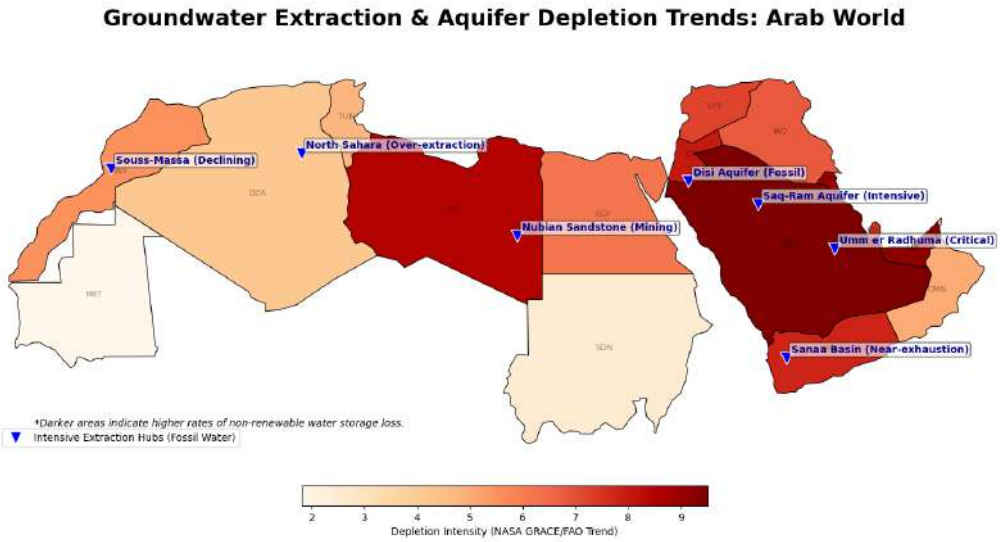


Figure 3.4: Groundwater Extraction and Aquifer Depletion Trends

3.2 Socio-Economic Classification

Traditional Subsistence Oases: These systems maintain traditional agricultural practices and water management institutions, with production primarily oriented toward local consumption and limited market exchange. They are characterized by strong social cohesion, traditional knowledge transmission, and limited integration into market economies. Examples include remote mountain oases in Oman and Yemen where traditional practices remain dominant.

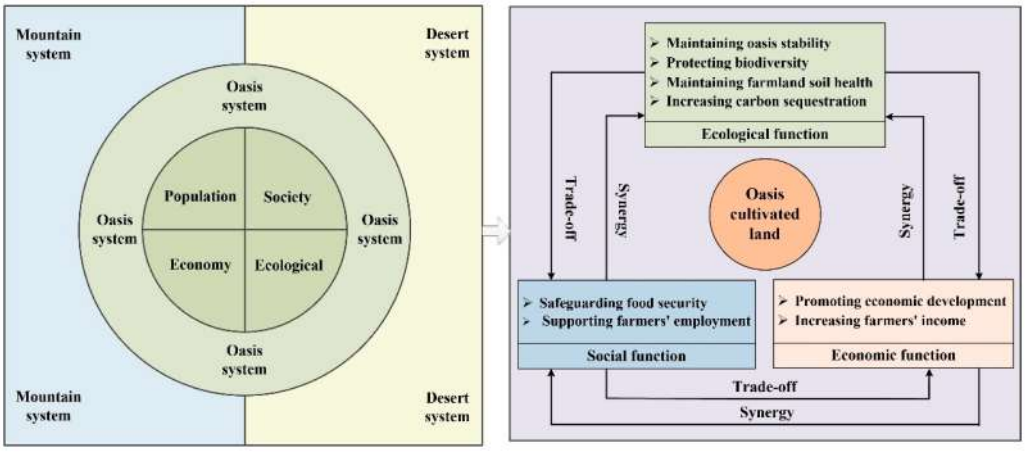


Figure 3.5: Socio-Economic Characteristics and Livelihood Systems in Oases

Transitioning Oases: These systems are experiencing significant socio-economic transformation, with traditional practices coexisting alongside modern technologies and market-oriented production. They often exhibit tensions between traditional and modern institutions, changing labor patterns, and evolving land tenure arrangements. Many oases in Morocco, Tunisia, and Egypt fall into this category [6].

Commercial Agricultural Oases: These systems are primarily oriented toward commercial agricultural production, often specializing in cash crops such as dates, fruits, or vegetables. They typically rely on modern irrigation technologies, market inputs, and wage labor. The hydrological characteristics often include intensive groundwater extraction and associated sustainability challenges. Examples include date palm oases in Saudi Arabia and the UAE.

Urban-Periurban Oases: These systems are integrated into or adjacent to urban areas, serving multiple functions including agricultural production, recreation, cultural heritage, and urban green space. They face unique pressures from urban expansion, land use competition, and changing water demands. Examples include oases near urban centers in Egypt, Jordan, and Gulf countries.

3.3 Development Trajectory Classification

Degraded Systems: These oases have experienced significant declines in water availability, soil quality, and agricultural productivity, often leading to outmigration and abandonment of traditional practices. The Azraq Oasis in Jordan represents a severely degraded system where natural springs have dried up and wetland ecosystems have collapsed.

Stable Traditional Systems: These oases have maintained traditional practices and institutions while adapting to changing conditions. They often have strong governance structures and balanced resource use. Some Aflaj systems in Oman and khettaras in Morocco demonstrate this stability.

Modernizing Systems: These oases are undergoing significant transformation through infrastructure development, technological adoption, and integration into market economies. They face challenges in balancing development with sustainability. The New Valley oases in Egypt exemplify this trajectory [5].

Restored Systems: These are oases that have experienced degradation but are undergoing restoration through integrated interventions. The Siwa Oasis in Egypt represents a partially restored system where interventions have addressed waterlogging but face ongoing challenges with governance and sustainability [4].

This typology provides a framework for understanding the diversity of oasis systems and developing targeted management strategies appropriate to their specific characteristics and challenges. It highlights the need for context-specific approaches that recognize the unique hydrological, socio-economic, and developmental conditions of each oasis type.

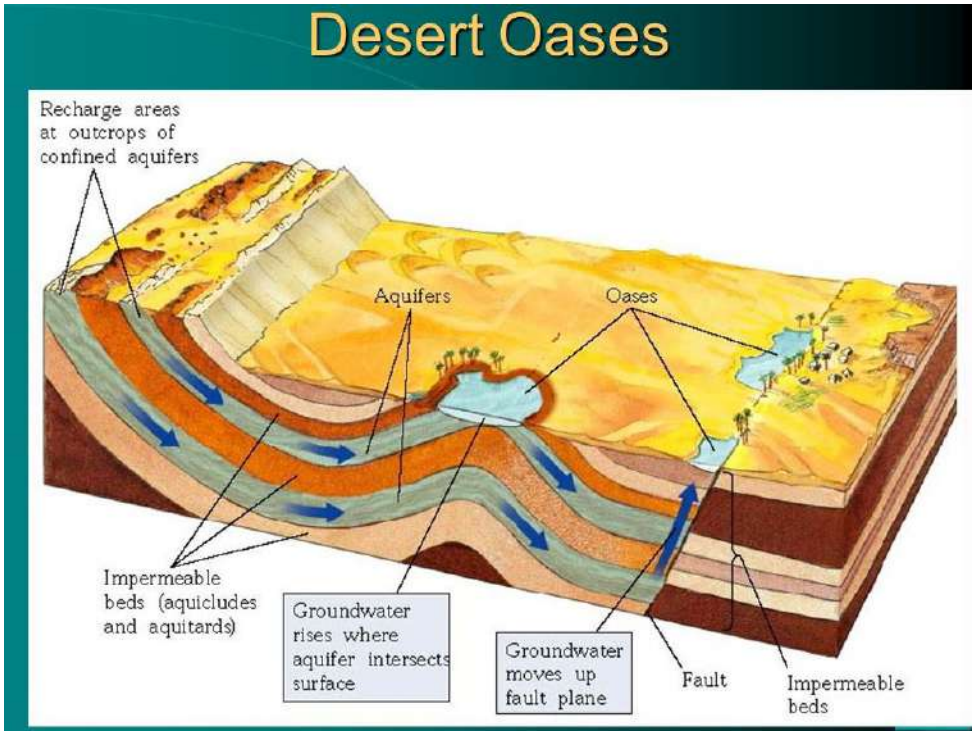


Figure 3.6: Typology of Oasis Structures and Functions

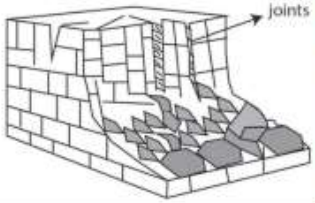
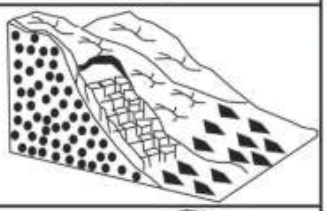
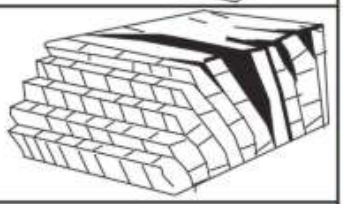
rock fall types	Natural causes of rock falls	Descriptive pattern	Field view
Rock Falls	Rock falls represent fractured and jointed rocks characterized by high steep slopes, resulting from stress release, joint opening, and hydrofracturing decrease (Stoffel etal. 2024). Individual blocks and rock avalanches are the products of rock falls. The joint development, high water pressure or steep slope are the main causes of the dislocation of the rock masses. Moreover, rushing climate warming and frozen degradation may be resulted in improved activity and large volume of blocks fall (Chiarle et al. 2021).Such natural feature is well observed in Taref Fm.		
Rock slumps	Slump is a category of mass wasting, owing to sliding of comprehensible rock material along a curved surface. Slump structures are well-known geological marvels on subaerial and submarine slopes that vary from a few tens of centimenters to several thusand cubic kilometers (Maltnan 1994). Gliding unconsodiated sediments and lithified blocks are outputs of this process (Martinsen and Bakken 1990). The latter has an imperative impact on human life known as catastrophe/disaster.		
Toppling	Rock creep is a slow downhill movement of rock debris that can be observed on a cliff face.During rock creeping, large jointed rocks and blocks slowly creep uniti they become unstable and topple over or side downhill (Bloom 1998).Surface creep is the result of fractures developing across small joint block near the ground surface and then the tilting downslope of the separated block (Burbank 2005).		
Rock slides	A rock slide is the sliding motion of rock along a sloping surface. Slides have predominantly straight pr shear planes and the sliding masses usually breakup into many blocks as it moves. In large rock slides, the shear planes may be stepped at major joints intersections with the shear planes and the rock mass may be broken up to become debris slide (Selby 1993; Valdiya 1998).		

Figure 3.7: Innovative Water Management Technologies for Oasis Sustainability

Figure 3.8: Science-Based Climate Solutions for Oases in the Arab Region

Chapter 4

A Comparative Analysis of Case Studies

A detailed examination of specific oasis systems across the Arab region reveals critical insights into the factors contributing to success or failure in sustainable management. The following comparative analysis synthesizes hydrological, institutional, and socio-economic dimensions of five major oasis systems:

Table 4.1: Comparative Analysis of Oasis Case Studies

Oasis Case Study	Hydrogeological Context	Primary Anthropogenic Pressures	Key Governance/Technical Interventions	Outcome & Critical Lesson Learned
Siwa Oasis, Egypt	Shallow aquifer with high salinity; historically supported by 150 natural springs with complex drainage patterns. Water table varies seasonally, creating natural lakes that fluctuate with evaporation rates.	Introduction of low-cost drilling technology in the 1980s led to uncontrolled proliferation of private wells (over 1,265 wells in less than 20 years). This disrupted the traditional communal spring system, causing waterlogging and salinization of thousands of acres of agricultural land.	Technical intervention in the 1990s to control "critical wells" and restore older irrigation systems. The plan involved decommissioning poorly designed wells and drilling new wells with control valves and water distribution systems. This reduced water losses to Siwa lake by 50% and restored lake dimensions to 1929 levels.	Initial technical success was not sustained due to lack of governance, monitoring, and community enforcement. Illegal drilling resumed, and agricultural expansion using available water led to renewed waterlogging. The critical lesson is that technological interventions without sustained institutional support and community engagement are insufficient for long-term sustainability.

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Table 4.1 – *Continued from previous page*

Oasis Case Study	Hydrogeological Context	Primary Anthropogenic Pressures	Key Governance/Technical Interventions	Outcome & Critical Lesson Learned
Azraq Oasis, Jordan	Located in the downstream part of a transboundary aquifer shared with Syria and Saudi Arabia. Historically supported by four major springs with a combined discharge of 15 million cubic meters annually, creating a critical wetland ecosystem.	Massive overpumping for Amman's water supply beginning in the 1980s, with extraction rates exceeding 40 million cubic meters annually. This, combined with upstream dam construction that reduced recharge, led to complete drying of natural springs by 1993 and a 90% reduction in wetland area.	Top-down conservation efforts initiated in the mid-1990s, including artificially pumping water back into the wetlands. These efforts were designed and implemented by government agencies with limited community involvement, focusing primarily on ecological restoration without addressing socio-economic dimensions.	Only partial restoration of approximately 10% of the original wetland area. The intervention created conflict with local communities who felt alienated from the process and excluded from decision-making. The critical lesson is that top-down restoration approaches that ignore local knowledge and community agency are socially and ecologically fragile, leading to limited long-term success.

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Table 4.1 – *Continued from previous page*

Oasis Case Study	Hydrogeological Context	Primary Anthropogenic Pressures	Key Governance/Technical Interventions	Outcome & Critical Lesson Learned
Aflaj Systems, Oman	Gravity-fed tunnel systems tapping shallow aquifers, with some systems over 2,500 years old. Water flow is maintained by gentle gradients, with distribution networks designed to minimize evaporation and seepage losses.	Breakdown of traditional social structures responsible for system maintenance, particularly the "Waqf" endowment system that funded maintenance. Modernization has led to reduced community participation, while government-provided alternatives have diminished the perceived value of traditional systems.	Millennia-old system of water law and precise allocation mechanisms using time units measured by sundials and stars. Water distribution follows a strict hierarchy based on quality and use, with drinking water of highest quality allocated first, followed by water for mosques, washing, and finally agriculture.	Systems are degrading where social capital has eroded, but remain functional where traditional institutions are maintained. The critical lesson is that the social and hydrological systems are inseparable; the collapse of community governance guarantees the collapse of the water system. Sustainable management requires maintaining both the physical infrastructure and the social institutions that govern it.

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Table 4.1 – *Continued from previous page*

Oasis Case Study	Hydrogeological Context	Primary Anthropogenic Pressures	Key Governance/Technical Interventions	Outcome & Critical Lesson Learned
Draa Valley, Morocco	Integrated system relying on both surface water from the El Mansour Eddahbi dam and groundwater from shallow aquifers. The system is characterized by extreme climate variability, with periodic droughts creating significant stress on water resources.	Over-exploitation of groundwater during drought periods, reaching 1.1 billion cubic meters annually. This has led to alarming declines in water levels—up to 4.5 meters per year in some aquifers—and widespread nitrate pollution from agricultural intensification.	Strong national legal framework (Law 36-15 on water) providing for participatory management contracts. Introduction of "groundwater contracts" as social pacts bringing together all water users to agree on sustainable use rules. Implementation of smart metering, water police, and mandatory data sharing between ministries.	Improved water management with better consideration of crop water requirements and more efficient dam releases. The critical lesson is that high-level political will, strong institutional tools, clear data, and inclusive governance structures are non-negotiable for effective groundwater management in complex socio-ecological systems.

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Table 4.1 – *Continued from previous page*

Oasis Case Study	Hydrogeological Context	Primary Anthropogenic Pressures	Key Governance/Technical Interventions	Outcome & Critical Lesson Learned
New Valley (Kharga, Dakhla, Farafra), Egypt	Dependent on the non-renewable Nubian Sandstone Aquifer System (NSAS), with historical artesian flow from deep wells. The aquifer has limited recharge, making it essentially a fossil resource with finite availability.	State-led agricultural expansion since the 1960s, involving drilling of hundreds of deep wells to support large-scale irrigation projects. Internal migration from the Nile Valley increased demand, leading to rapid decline in natural spring flow within a decade and continuous decreases in groundwater levels.	Large-scale irrigation projects and groundwater monitoring programs. Technical solutions to address waterlogging and salinization through drainage systems. Development of groundwater management plans attempting to balance extraction with sustainable yield.	Rapid hydrological changes with declining water levels and deteriorating water quality. The critical lesson is that large-scale development in non-renewable aquifer systems requires disaggregated management plans tailored to local hydrogeological and socioeconomic conditions, rather than uniform policies applied across diverse contexts.

4.1 Cross-Cutting Analysis

This comparative analysis reveals several critical patterns that transcend individual cases:

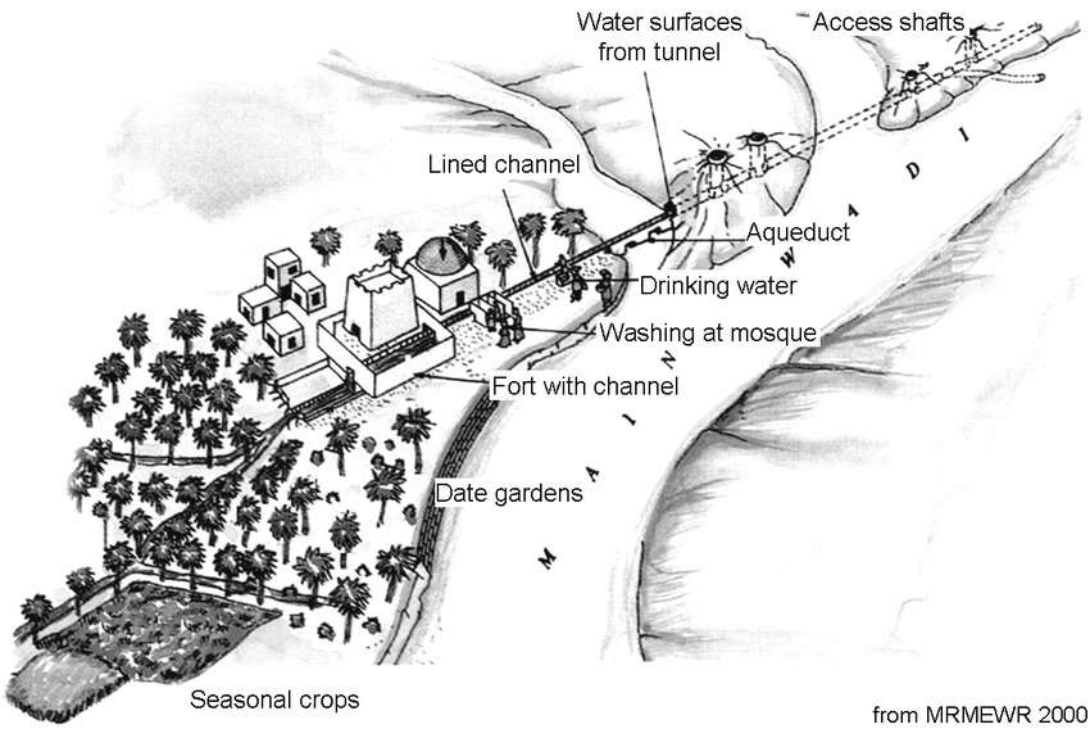


Figure 4.1: Traditional Aflaj Water Management System in Oman: Structure and Distribution

The Technology-Governance Nexus: Technological interventions without corresponding governance structures consistently fail to achieve long-term sustainability. The Siwa case demonstrates that even well-designed technical solutions will deteriorate without ongoing monitoring, enforcement, and community engagement [4]. Conversely, the Draa Valley case shows how strong governance frameworks can enhance the effectiveness of both traditional and modern technologies.

The Community Participation Imperative: Cases that successfully engage local communities as active participants rather than passive beneficiaries show greater resilience and sustainability. The Aflaj systems in Oman have persisted for millennia precisely because they are embedded in strong social institutions that ensure maintenance and equitable distribution. In contrast, the top-down approach in Azraq created conflict and limited ecological outcomes.

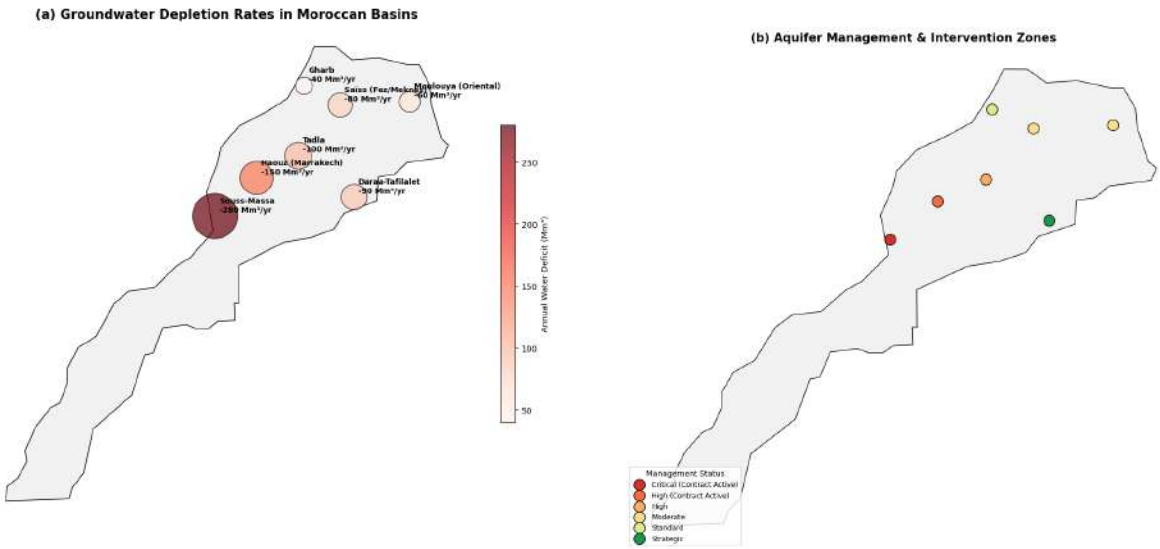
The Integration Challenge: The most promising approaches combine traditional knowledge with modern science and technology. The Draa Valley case demonstrates how traditional systems like the khettara (underground tunnels) and seguia (open-air canals) can be integrated with modern infrastructure like dams and smart metering to create resilient hybrid systems.

The Economic Viability Factor: Traditional practices face increasing pressure from economic changes that make them less viable for younger generations. Across all cases,

Figure 4.2: Water Resources Management in Morocco: National Strategy and Planning Framework

the economic marginalization of traditional agriculture emerges as a critical threat, driving youth migration and abandonment of sustainable practices. Successful approaches must address this economic dimension by creating viable livelihoods based on sustainable practices.

The Policy Coherence Challenge: Fragmented policies across water, agriculture, environment, and cultural heritage sectors undermine sustainable management. Cases with more integrated policy frameworks, such as Morocco's comprehensive water law, demonstrate better outcomes than those with sectoral approaches.



(a) Groundwater Depletion Rates (b) Aquifer Management Zones

Figure 4.3: Groundwater Resources Assessment in Morocco

This analysis underscores that sustainable oasis management requires a holistic approach that addresses hydrological, ecological, socio-economic, and governance dimensions simultaneously. It highlights the need for context-specific strategies that build on local knowledge and institutions while incorporating appropriate modern technologies and governance mechanisms.

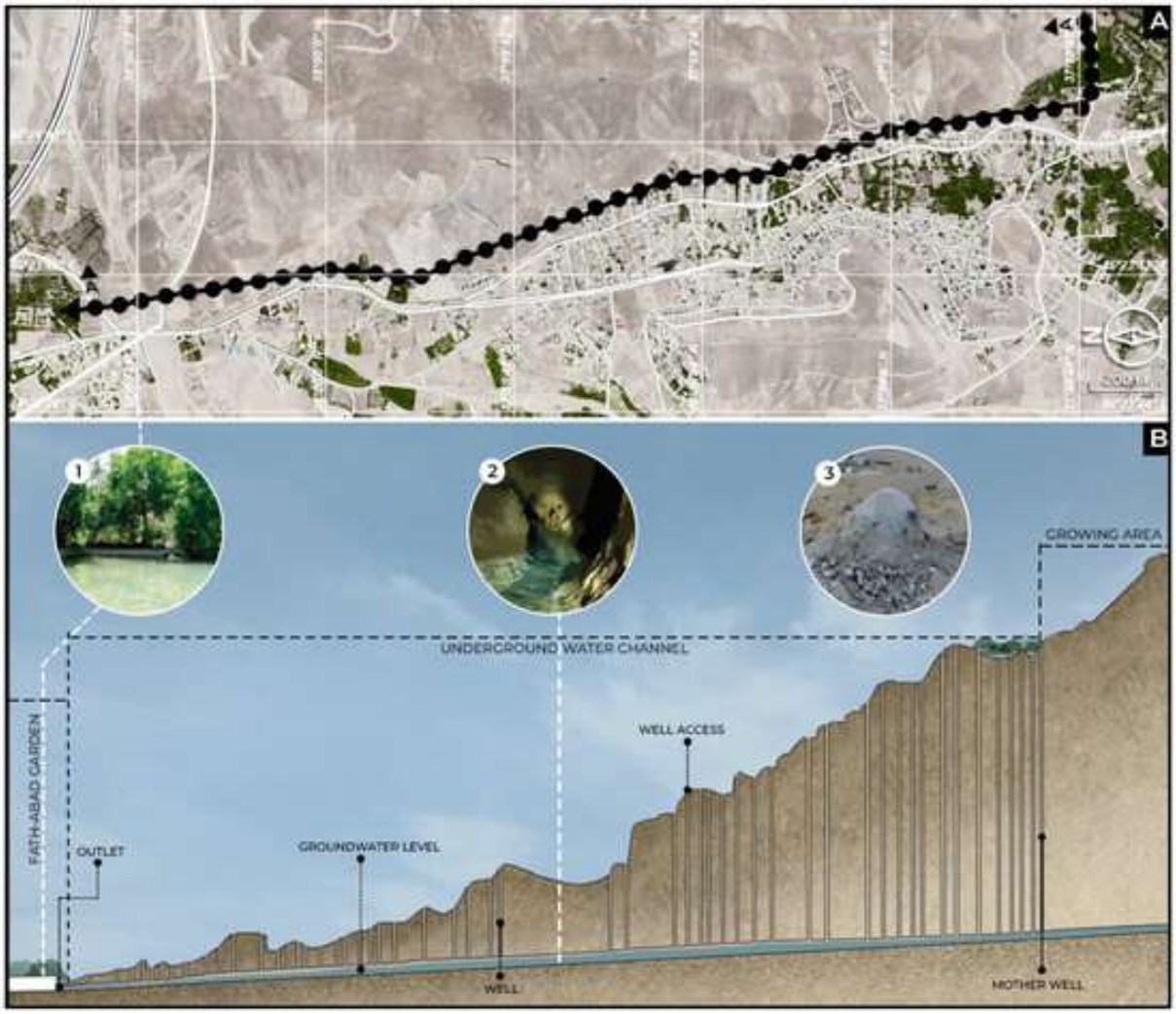


Figure 4.4: Traditional Hydraulic Systems in Moroccan Oases: Heritage Assets

Figure 4.5: Draa Valley Oasis: Water Resources Management and Agricultural Systems

Chapter 5

Systemic Challenges and Strengths - A Consolidated SWOT Analysis

A comprehensive assessment of oasis systems in the Arab region reveals a complex landscape of challenges and opportunities. This consolidated SWOT analysis synthesizes insights from expert analyses and case studies to provide a strategic overview of the current situation:

5.1 Strengths

Sophisticated Traditional Knowledge Systems: Oasis communities possess detailed knowledge of local hydrology, microclimates, soil types, and plant varieties that has been accumulated and refined over generations. This knowledge includes sophisticated water management techniques, agricultural practices, and architectural designs that are well-adapted to arid environments. The precision of traditional water allocation systems in the Omani Aflaj, for instance, demonstrates a level of hydrological understanding that rivals modern scientific approaches.

Genetic Resources and Agrobiodiversity: Oases harbor unique genetic resources, particularly of date palms and other fruit trees, that have been selected and preserved for centuries. These varieties possess valuable traits such as drought tolerance, salinity resistance, and specific nutritional qualities. The diversity of date palm varieties in traditional oases represents an invaluable genetic resource for climate change adaptation and food security.

Proven Resilience: Many oasis systems have demonstrated remarkable resilience over centuries or even millennia, adapting to climate variations, social changes, and technological innovations while maintaining their core functions. This resilience is embedded in the flexible institutions, diversified livelihoods, and adaptive management practices that characterize traditional oasis systems.

Existing Institutional Frameworks: Several countries have established institutional mechanisms for oasis management, such as Morocco's National Agency for the Development of Oasis and Argan Zones (ANDZOA) and Oman's Ministry of Regional Municipalities and Water Resources. These institutions provide platforms for coordination, policy

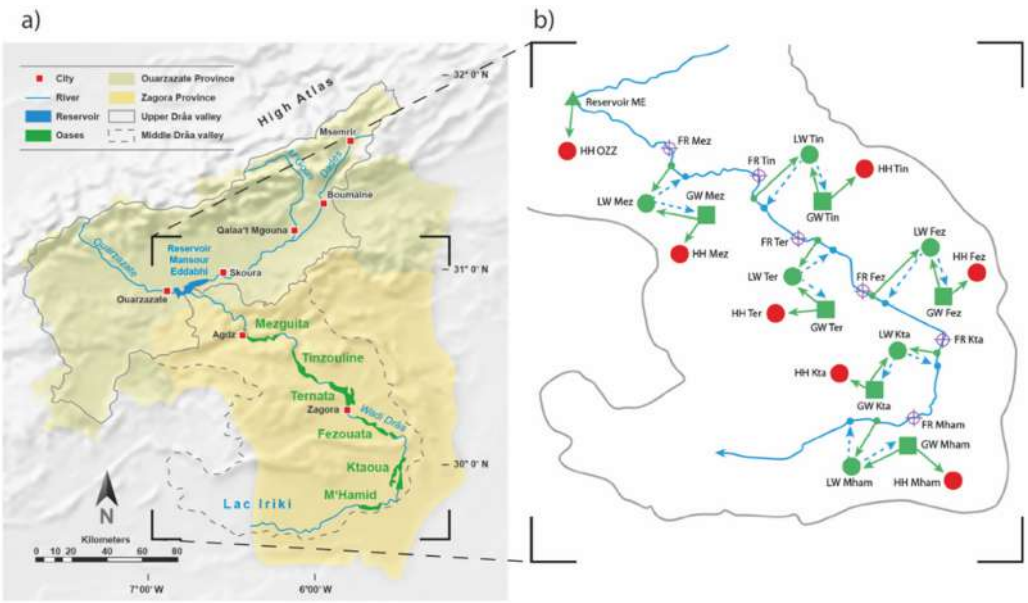


Figure 5.1: Draa Valley Water Management: Integrated Surface and Groundwater Systems

development, and implementation of conservation measures.

International Recognition and Support: UNESCO's involvement through programs like the Man and the Biosphere (MAB) Programme and World Heritage designation provides international visibility, technical support, and funding opportunities for oasis conservation [3]. These designations also create incentives for national governments to invest in conservation and sustainable development.

5.2 Weaknesses

Over-reliance on Finite Resources: Many oasis systems depend on non-renewable groundwater aquifers that are being depleted at unsustainable rates. The Nubian Sandstone Aquifer System, which supports the New Valley oases in Egypt, has finite reserves that are being extracted far faster than they can be replenished, creating a fundamental sustainability challenge [5].

Fragmented Management Approaches: Management responsibilities are often fragmented across multiple agencies and sectors, with limited coordination between water, agriculture, environment, and cultural heritage authorities. This fragmentation leads to conflicting policies, duplicated efforts, and gaps in coverage that undermine effective management.

Loss of Traditional Knowledge: The transmission of traditional knowledge is being disrupted by social changes, outmigration of youth, and the declining economic viability of traditional practices. As elders pass away without transferring their knowledge to younger generations, critical management expertise is being lost, creating a knowledge gap that cannot be easily filled by modern science alone.

Limited Monitoring and Enforcement Capacity: Many countries lack the technical capacity and resources for comprehensive monitoring of groundwater resources, ecosystem

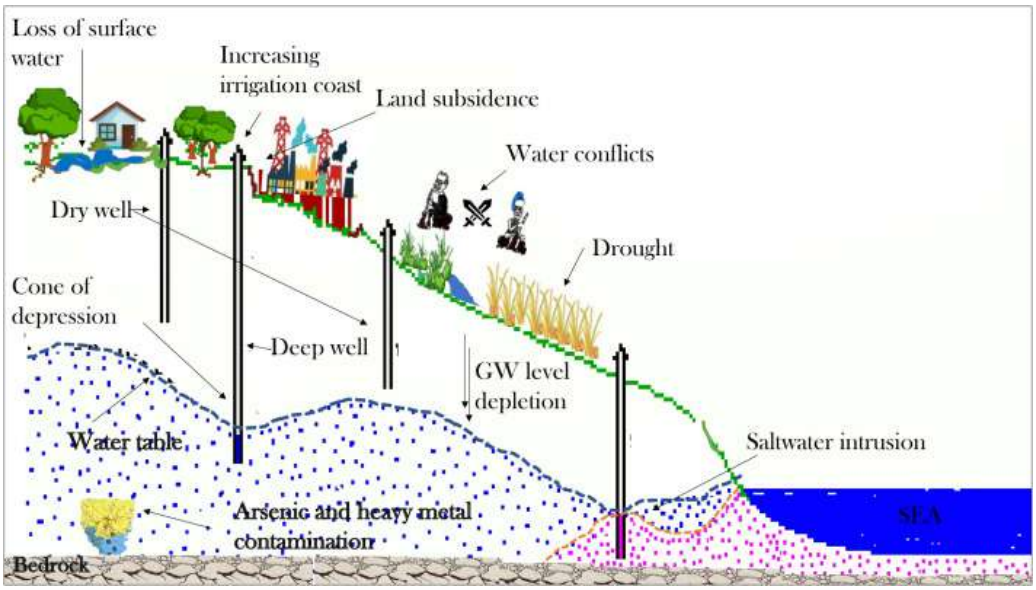


Figure 5.2: Groundwater Depletion and Management Challenges in Oasis Systems

health, and compliance with regulations. This limitation undermines effective management, as decisions cannot be based on accurate information, and violations cannot be detected and addressed in a timely manner.

Economic Marginalization: Traditional oasis agriculture is often economically marginal compared to alternative livelihoods, leading to abandonment of sustainable practices and outmigration of youth. This economic pressure creates a vicious cycle where declining human capacity leads to deteriorating resource conditions, further reducing economic viability.

5.3 Opportunities

Climate Finance for Adaptation: The growing availability of climate finance for adaptation measures presents an opportunity to invest in oasis conservation and sustainable development [7]. By positioning oases as models for climate-resilient agriculture and water management, countries can access international funding to support conservation efforts.

Premium Markets for Heritage Products: Growing consumer interest in authentic, sustainable, and culturally significant products creates market opportunities for oasis communities. Certified heritage products, particularly date palms and other traditional crops, can command premium prices that provide economic incentives for conservation.

Technological Innovations: New technologies offer opportunities to enhance water efficiency, monitoring capabilities, and agricultural productivity. Remote sensing can provide real-time data on vegetation health and water use, while precision irrigation technologies can reduce water losses and improve crop yields.

Regional Cooperation Mechanisms: Existing regional bodies such as the Arab Ministerial Council on Water and the Arab Water Council provide platforms for regional cooperation on oasis management. A coordinated regional approach could facilitate knowledge sharing, harmonize policies, and leverage resources for conservation.

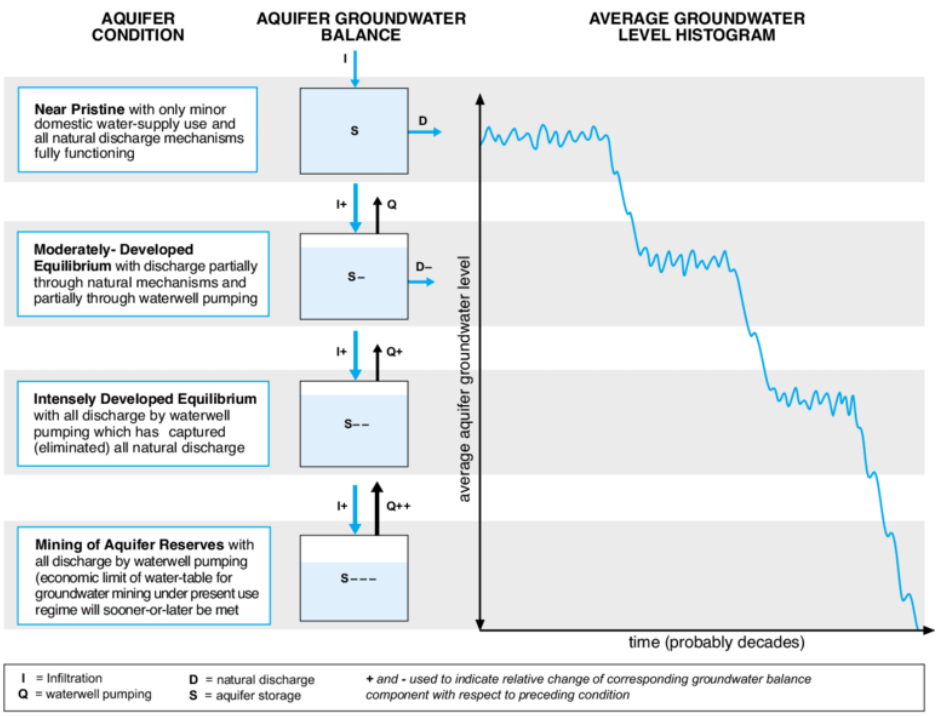


Figure 5.3: New Valley Oases: Groundwater Resources and Development Trajectories

Integration with Sustainable Development Goals: Oases contribute to multiple Sustainable Development Goals, including zero hunger (SDG 2), clean water and sanitation (SDG 6), sustainable cities and communities (SDG 11), climate action (SDG 13), and life on land (SDG 15). This alignment creates opportunities to integrate oasis conservation into national development strategies and access associated funding streams.

5.4 Threats

Climate Change Impacts: Climate change is exacerbating water scarcity through increased temperatures, reduced precipitation, and more frequent extreme weather events [1]. These changes directly impact groundwater recharge rates, agricultural productivity, and ecosystem health, creating additional stress on already vulnerable systems.

Unsustainable Subsidies: Government subsidies for water, energy, and agricultural inputs often create perverse incentives that encourage over-exploitation of resources. Subsidized electricity for groundwater pumping, for instance, has led to excessive extraction in many oasis systems, while fertilizer subsidies have contributed to water pollution.

Rapid Urbanization and Land Use Change: Expanding urban areas and infrastructure development encroach on oasis ecosystems and compete for water and land resources. This pressure is particularly acute in peri-urban oases, where urban expansion often leads to the conversion of agricultural land and disruption of traditional water systems.

Socio-Economic Transformations: Changing economic structures, demographic shifts, and evolving cultural values are transforming traditional oasis societies. These changes include youth outmigration, declining agricultural labor force, and shifting consumption patterns that reduce demand for traditional products and practices.

Political Instability and Conflict: Regional instability and conflict can disrupt management efforts, divert attention and resources from conservation, and create refugee flows that place additional pressure on oasis resources. Political instability also undermines long-term planning and investment in sustainable management.

The Causal Chain of Subsidy-Driven Ecosystem Collapse

A critical threat requiring detailed analysis is the role of unsustainable subsidies in driving ecosystem degradation. This process follows a predictable causal chain:

1. **Subsidy Introduction:** Governments introduce subsidies for water, energy, or agricultural inputs with the intention of supporting rural livelihoods and increasing agricultural production.
2. **Reduced Resource Costs:** Subsidies lower the apparent cost of resource extraction and use, making it economically viable to extract water from greater depths, apply more fertilizers, or expand irrigated areas.
3. **Increased Extraction and Use:** Lower costs lead to increased resource extraction beyond sustainable levels. In the case of groundwater, this means pumping rates exceed recharge rates, leading to declining water tables.
4. **Ecosystem Degradation:** Over-extraction leads to ecosystem degradation, including declining water quality, soil salinization, and loss of biodiversity. These changes reduce the productivity and resilience of the system.
5. **Increased Dependency:** As degradation progresses, systems become less productive and more vulnerable to external shocks, increasing dependency on subsidies and creating a vicious cycle of degradation and dependency.
6. **System Collapse:** In extreme cases, this process can lead to complete system collapse, where ecosystems and livelihoods become non-viable even with continued subsidies.

This causal chain has been observed in multiple contexts, from traditional grazing systems to modern groundwater extraction. The historical example of the Nizam al-Hima system in Saudi Arabia, where government subsidies for barley led to herd sizes exploding beyond the land's carrying capacity, resulting in total ecosystem degradation, illustrates this process clearly. Similar patterns are evident in many oasis systems where subsidized electricity for groundwater pumping has led to excessive extraction and aquifer depletion.

This SWOT analysis reveals that while oases face significant challenges, they also possess unique strengths and opportunities that can be leveraged for their sustainable management. The key to success lies in addressing the weaknesses and threats while building on the inherent strengths and opportunities through integrated, context-specific strategies.

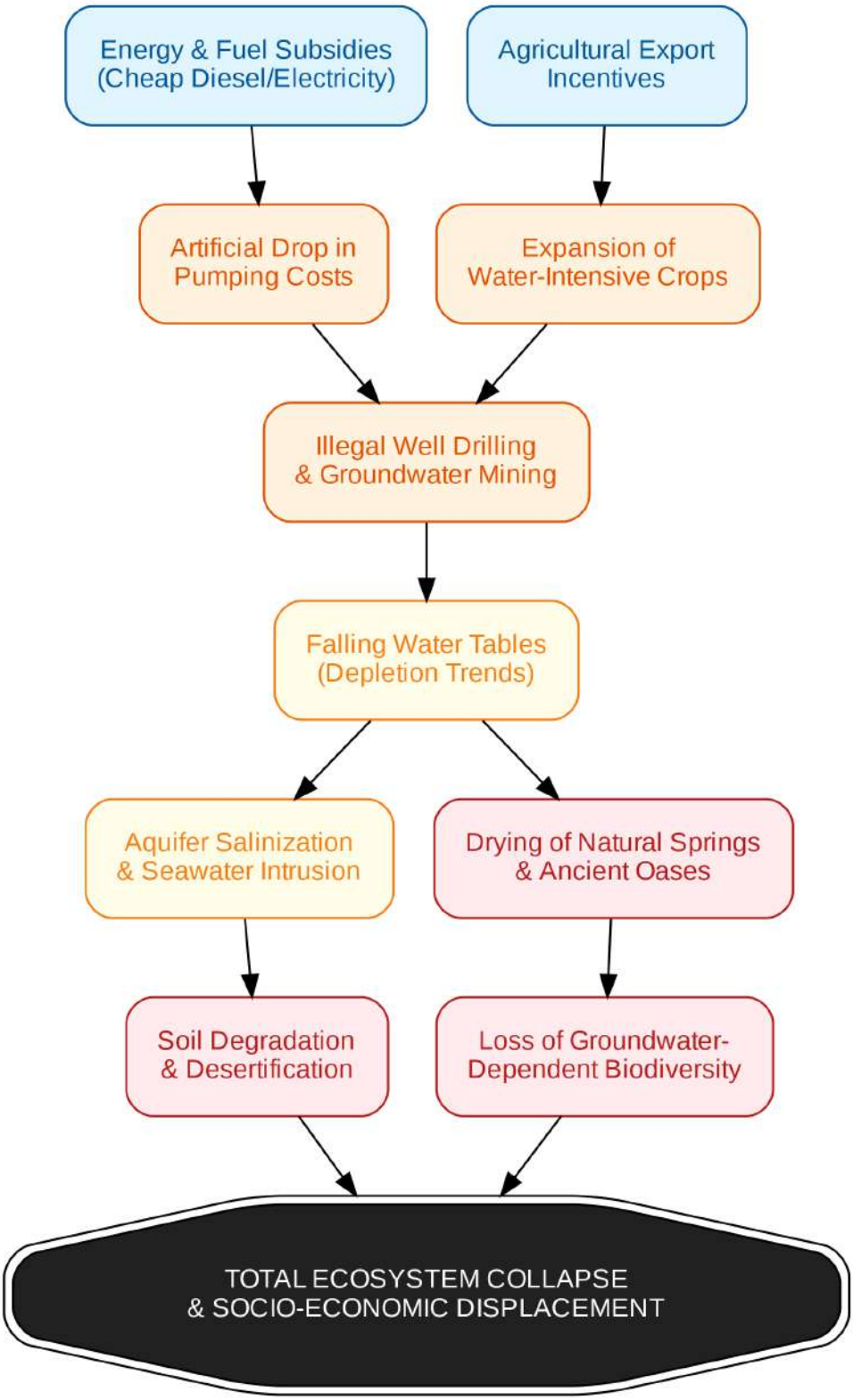


Figure 5.4: Causal Chain Diagram of Subsidy-Driven Ecosystem Collapse

Figure 5.5: New Valley Oasis: Field Assessment and Community Engagement

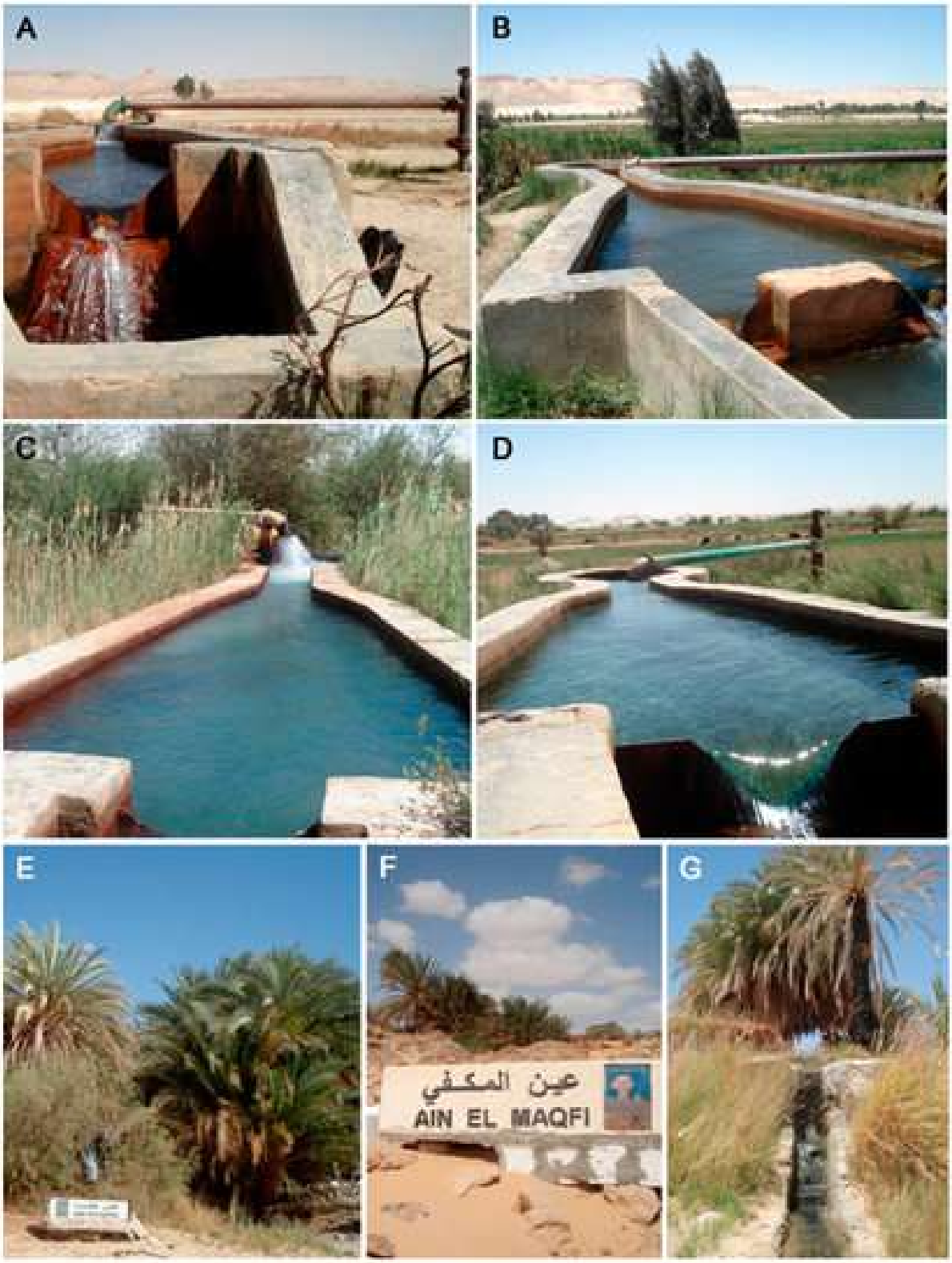


Figure 5.6: Traditional Water Infrastructure in New Valley Oases

Figure 5.7: Agricultural Systems and Date Palm Cultivation in New Valley

Figure 5.8: Community Water Management Practices in New Valley Oases

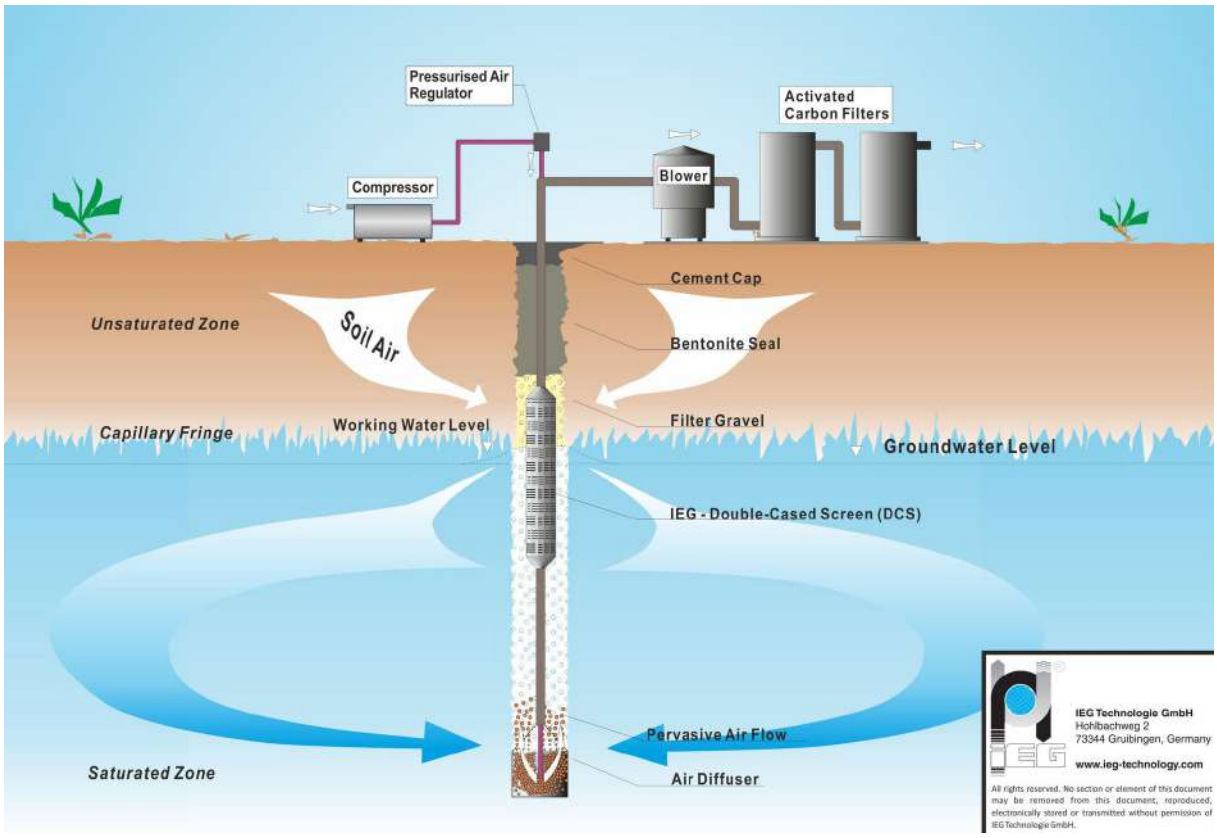


Figure 5.9: Groundwater Wells and Extraction Systems in New Valley

Part III: The Prescription – A Strategic Framework and Action Plan for Resilience

Chapter 6

The Four Pillars of Sustainable Oasis Management

The diagnosis of oasis systems reveals the need for a comprehensive strategic framework that addresses the complex interplay of hydrological, ecological, socio-economic, and governance dimensions. This framework is built upon four integrated pillars that reflect the paradigm shift toward heritage-based solutions:

6.1 Pillar 1: An Integrated and Context-Specific Approach to Water Management

This pillar rejects uniform, one-size-fits-all approaches in favor of integrated strategies that are tailored to local hydrogeological conditions, socio-economic contexts, and cultural traditions. It emphasizes the need to combine modern engineering solutions with traditional knowledge systems, recognizing that each approach has unique strengths and limitations.

Modern Engineering Solutions: Modern technologies offer valuable tools for enhancing water management in oasis systems. Artificial groundwater recharge, for instance, can significantly increase the effective storage capacity of aquifers in arid regions. Research in the Gulf indicates that with rechargeable wells, the amount of stored water can reach 85% of precipitation, compared to only 15% through natural recharge. Similarly, precision irrigation technologies such as subsurface drip irrigation can reduce evaporation losses and improve water use efficiency by 30-50% compared to traditional surface irrigation methods.

However, the applicability of these technologies depends on specific local conditions. Artificial recharge, for example, requires surplus water (from flash floods or treated wastewater) and suitable geological conditions for infiltration. In Egypt, where such surplus is limited, the primary method has been the construction of dikes across wadis to slow floodwaters, but this approach has led to increased soil salinization as standing water evaporates, leaving behind concentrated salts.

Traditional Water Management Systems: Traditional systems such as the Aflaj of Oman, Qanats of Iran, and Khettaras of Morocco represent sophisticated engineering

achievements that have sustained oasis systems for centuries or even millennia. These gravity-flow systems minimize energy requirements, reduce evaporation losses, and provide consistent water flow through careful design and maintenance. The Aflaj system, for instance, incorporates detailed understanding of groundwater hydrology, tunnel engineering, and water distribution that enables sustainable extraction from shallow aquifers.

The restoration and maintenance of these systems offer significant potential for enhancing water security in oasis regions. However, their effectiveness depends on the social institutions that govern them, including water allocation mechanisms, maintenance responsibilities, and conflict resolution processes. Where these institutions have eroded, traditional systems have often fallen into disrepair, even when the physical infrastructure remains intact.

Integrated Approaches: The most promising strategies combine traditional and modern approaches in ways that leverage the strengths of each. In the Draa Valley of Morocco, for instance, traditional khettaras and seguia systems are integrated with modern dam infrastructure and smart metering to create a hybrid system that optimizes water use across different sources and users. Similarly, in some Omani Aflaj systems, modern monitoring technologies are being used to complement traditional management practices, providing real-time data on water flow and quality while maintaining traditional allocation mechanisms.

The key to successful integration is ensuring that modern technologies complement rather than replace traditional knowledge and institutions. This requires careful assessment of local conditions, participatory planning processes, and adaptive management approaches that can respond to changing conditions and new information.

6.2 Pillar 2: A Foundational Inventory and Typology of Oases

A critical barrier to effective oasis management is the lack of comprehensive, standardized data on the distribution, characteristics, and status of oasis systems across the Arab region. This pillar addresses this gap through the development of a foundational inventory and typology that can inform targeted planning and management strategies.

Inventory Development: A comprehensive oasis inventory would include detailed mapping of oasis locations, boundaries, and key characteristics, including:

- **Hydrological Characteristics:** Water sources, aquifer types, recharge mechanisms, water quality, and extraction rates
- **Ecological Characteristics:** Vegetation types, biodiversity indicators, land use patterns, and ecosystem services
- **Socio-Economic Characteristics:** Population demographics, economic activities, land tenure systems, and institutional arrangements
- **Cultural Characteristics:** Traditional knowledge systems, cultural practices, heritage sites, and intangible cultural heritage

This inventory would be developed through a combination of remote sensing, field surveys, and participatory mapping involving local communities. Remote sensing technologies,

including satellite imagery and drone surveys, can provide baseline data on land cover, vegetation health, and water use patterns, while field surveys can ground-truth this data and collect more detailed information on hydrological and socio-economic conditions.

Typology Development: Building on the inventory, a multi-dimensional typology would classify oases according to their hydrogeological, socio-economic, and functional characteristics, as outlined in Chapter 3. This typology would enable:

- **Targeted Management Strategies:** Development of management approaches appropriate to specific oasis types, recognizing that different systems face distinct challenges and opportunities
- **Comparative Analysis:** Identification of patterns and trends across different types of oases, facilitating knowledge sharing and transfer of successful approaches
- **Policy Development:** Formulation of policies that address the specific needs of different oasis types while maintaining overall coherence and consistency

Implementation Framework: The inventory and typology would be developed through a phased approach, beginning with pilot studies in representative oasis systems across the region, followed by scaling up to cover the entire Arab region. A regional coordinating body would oversee standardization of methodologies, quality assurance, and data sharing, while national agencies would be responsible for data collection and management at the country level.

The inventory and typology would be maintained as dynamic systems, with regular updates to reflect changing conditions and new information. A digital platform would provide access to the data for researchers, policymakers, and local communities, with appropriate safeguards for sensitive information and indigenous knowledge.

6.3 Pillar 3: The Centrality of Indigenous Knowledge and Community Empowerment

This pillar represents a fundamental shift from conventional top-down development approaches to ones that recognize local communities as the primary agents and experts in oasis management. It emphasizes the need to empower communities through legal recognition, capacity building, and economic incentives that make traditional practices viable in contemporary contexts.

Legal Recognition of Traditional Institutions: Traditional water management institutions, such as water user associations, irrigation cooperatives, and customary governance structures, play critical roles in sustainable resource management. However, these institutions often lack formal legal recognition, limiting their authority and capacity to enforce rules and regulations.

Legal recognition would involve formalizing the status of these institutions, defining their rights and responsibilities, and establishing mechanisms for coordination with government agencies. In Morocco, for instance, water user associations have been formally recognized and given responsibilities for water distribution and conflict resolution within their areas, enhancing the effectiveness of both traditional and modern management systems.

Knowledge Documentation and Validation: Traditional knowledge systems are at risk of being lost as elders pass away without transmitting their expertise to younger generations. Systematic documentation and validation of this knowledge is therefore urgent.

Documentation efforts should capture not only the technical aspects of traditional practices but also the cultural contexts, institutional arrangements, and adaptive management strategies that have sustained these systems. Validation processes should involve collaboration between traditional knowledge holders and scientists, combining local observations with scientific analysis to create a more comprehensive understanding of oasis systems.

Economic Incentives for Traditional Practices: The economic viability of traditional practices is a critical factor in their continued transmission and adoption. Creating economic incentives for these practices involves:

- **Market Development:** Establishing markets and value chains for traditional products, particularly those with unique qualities or cultural significance
- **Certification Systems:** Developing certification schemes that recognize and reward sustainable production methods and cultural heritage values
- **Payment for Ecosystem Services:** Creating mechanisms to compensate communities for maintaining ecosystem services provided by traditional management practices
- **Ecotourism:** Developing tourism models that showcase traditional practices and provide direct economic benefits to communities

Capacity Building and Intergenerational Transmission: Ensuring the continuity of traditional knowledge requires investment in capacity building and intergenerational transmission. This involves:

- **Education Programs:** Integrating traditional knowledge into formal and informal education systems
- **Apprenticeship Programs:** Creating structured opportunities for youth to learn from elders through hands-on experience
- **Innovation Platforms:** Establishing forums where traditional knowledge holders and innovators can collaborate to adapt traditional practices to contemporary challenges

6.4 Pillar 4: Leveraging Science and Technology as an Enabler, Not a Replacement

This pillar addresses the need to harness the potential of science and technology to enhance the sustainability and resilience of oasis systems while ensuring that these tools complement rather than replace traditional knowledge and practices.

Appropriate Technology Selection: The selection of technologies for oasis management should be based on careful assessment of local conditions, needs, and capacities. Appropriate technologies are those that:

- Address Critical Needs: Target the most pressing challenges facing oasis communities
- Are Accessible and Affordable: Can be acquired, maintained, and operated with local resources and capacities
- Are Compatible with Local Contexts: Respect and complement traditional knowledge and practices
- Enhance Resilience: Increase the capacity of systems to withstand and recover from shocks and stresses

Examples of appropriate technologies include low-cost groundwater level sensors, solar-powered pumps for remote areas, mobile applications for water management, and low-cost water quality testing kits.

Monitoring and Early Warning Systems: Effective management depends on accurate, timely information about resource conditions and trends. Modern monitoring technologies can significantly enhance the capacity of oasis communities to manage their resources sustainably:

- **Remote Sensing:** Satellite imagery can provide data on vegetation health, water use, and land use changes at multiple scales
- **Sensor Networks:** Ground-based sensors can monitor groundwater levels, water quality, and meteorological conditions in real-time
- **Early Warning Systems:** Integration of monitoring data with predictive models can provide early warning of droughts, floods, and other extreme events

These technologies are most effective when combined with traditional monitoring approaches, such as indicator species and phenological observations, creating hybrid systems that leverage the strengths of both scientific and traditional knowledge.

Decision Support Systems: Complex water management decisions require integration of multiple types of data and consideration of various scenarios and trade-offs. Decision support systems can assist by:

- **Data Integration:** Combining hydrological, ecological, socio-economic, and institutional data into comprehensive assessments
- **Scenario Analysis:** Modeling the potential impacts of different management options under various conditions
- **Stakeholder Engagement:** Providing platforms for participatory decision-making that incorporate diverse perspectives and knowledge systems

Knowledge Management and Sharing: Effective use of science and technology requires robust systems for knowledge management and sharing. These systems should:

- **Integrate Multiple Knowledge Systems:** Combine scientific and traditional knowledge in ways that respect the integrity of each
- **Facilitate Access:** Ensure that knowledge is accessible to diverse stakeholders, including local communities, policymakers, and researchers

- **Support Learning:** Create mechanisms for continuous learning and adaptation based on experience and new information

Together, these four pillars provide a comprehensive framework for sustainable oasis management that addresses the complex challenges facing these systems while building on their inherent strengths and opportunities. The framework emphasizes the need for integrated, context-specific approaches that recognize the central role of communities and the value of traditional knowledge, while harnessing the potential of science and technology to enhance resilience and sustainability.

Chapter 7

Strategic Action Plan for Oasis MAB Reserves

Translating the strategic framework into concrete action requires a structured approach that addresses governance, knowledge management, and socio-economic viability. The following action plan provides specific, actionable recommendations for the sustainable management of oasis ecosystems, with particular attention to their integration into the UNESCO Man and the Biosphere (MAB) Programme [8].

7.1 Theme 1: Governance & Policy Integration

Action 1.1: Establish Permanent Regional Committee for Oasis Management

Objective: Create a permanent institutional mechanism for regional coordination, policy development, and resource mobilization for oasis conservation and sustainable development.

Implementation Steps:

1. Develop formal proposal for establishment of an "Oasis Management Sub-Committee" under the Arab Ministerial Council on Water, with representation from water, environment, agriculture, and cultural heritage sectors.
2. Define mandate, structure, and operational procedures for the committee, including provisions for technical working groups, stakeholder consultation, and reporting mechanisms.
3. Secure endorsement from League of Arab States and establish secretariat function within UNESCO Regional Office for Cairo.
4. Develop initial work program focusing on harmonization of policies, development of regional guidelines, and mobilization of international support.

Timeline: Proposal development by Q4 2025; establishment by Q2 2026; full operationalization by Q4 2026.

Responsible Actors: UNESCO Regional Office for Cairo, League of Arab States, Arab Water Council, national water and environment ministries.

Action 1.2: Integrate Oasis Strategies into National Adaptation Plans

Objective: Ensure that oasis-specific considerations are incorporated into national climate adaptation strategies and action plans.

Implementation Steps:

1. Develop guidelines and templates for oasis-specific vulnerability assessments and adaptation strategies.
2. Conduct pilot assessments in three representative countries (Egypt, Morocco, Oman) to test and refine methodologies.
3. Develop capacity building program for national agencies on oasis-specific adaptation planning.
4. Establish mechanism for monitoring and reporting on implementation of oasis-specific adaptation measures.

Timeline: Guidelines development by Q1 2026; pilot assessments completed by Q3 2026; capacity building program launched by Q4 2026.

Responsible Actors: National climate change authorities, UNESCO, UNEP, Arab Water Council, research institutions.

Action 1.3: Develop Legal Framework for Community-Based Water Governance

Objective: Create model legislation and regulatory frameworks that formally recognize and empower traditional water user associations and community-based governance structures.

Implementation Steps:

1. Review existing legal frameworks in selected countries to identify best practices and gaps.
2. Develop model legislation for recognition of water user associations, defining their rights, responsibilities, and relationship with government agencies.
3. Develop regulatory framework for community-based water management, including provisions for water allocation, conflict resolution, and infrastructure management.
4. Pilot implementation in three MAB reserves, with evaluation and refinement based on experience.

Timeline: Legal review completed by Q2 2026; model legislation developed by Q4 2026; pilot implementation initiated by Q2 2027.

Responsible Actors: National water ministries, legal experts, water user associations, UNESCO MAB Programme.

Action 1.4: Reform Subsidy Policies for Sustainable Resource Use

Objective: Reform subsidy policies that create perverse incentives for over-exploitation of water and other resources in oasis systems.

Implementation Steps:

1. Conduct comprehensive assessment of current subsidy policies and their impacts on resource use in oasis systems.
2. Develop policy options for restructuring subsidies to align with sustainable management objectives, including phased approaches and transition support.
3. Develop economic models to assess impacts of subsidy reforms on different stakeholder groups, particularly vulnerable households.
4. Pilot test reformed subsidy approaches in selected oasis systems, with monitoring and evaluation of outcomes.

Timeline: Assessment completed by Q3 2026; policy options developed by Q1 2027; pilot implementation initiated by Q3 2027.

Responsible Actors: Finance ministries, water and agriculture ministries, economic research institutions, international financial institutions.

7.2 Theme 2: Knowledge Management & Innovation

Action 2.1: Develop Regional Oasis Knowledge Platform

Objective: Create a comprehensive digital platform for sharing data, knowledge, and best practices on oasis management across the Arab region.

Implementation Steps:

1. Design platform architecture and data standards, ensuring compatibility with existing systems and interoperability across countries.
2. Develop core modules for oasis inventory, typology, case studies, traditional knowledge documentation, and monitoring data.
3. Implement data sharing protocols and governance mechanisms, including provisions for data sovereignty and protection of sensitive information.
4. Develop capacity building program for national agencies and local communities on platform use and contribution.

Timeline: Platform design completed by Q2 2026; core modules developed by Q4 2026; full operationalization by Q2 2027.

Responsible Actors: UNESCO Regional Office for Cairo, Arab Water Council, national research institutions, IT specialists.

Action 2.2: Establish Traditional Knowledge Documentation Program

Objective: Systematically document and validate traditional knowledge related to water management, agriculture, and biodiversity conservation in oasis systems.

Implementation Steps:

1. Develop methodologies for participatory documentation of traditional knowledge, including protocols for prior informed consent and benefit-sharing.
2. Establish national documentation teams with expertise in ethnobotany, hydrology, agronomy, and social sciences.
3. Implement documentation program in 20 representative oasis systems across the region, with particular focus on knowledge held by elders.
4. Develop validation processes involving collaboration between traditional knowledge holders and scientists, with publication of validated knowledge in accessible formats.

Timeline: Methodologies developed by Q3 2026; documentation teams established by Q4 2026; documentation completed in target sites by Q4 2027.

Responsible Actors: National research institutions, universities, traditional knowledge holders, UNESCO, local communities.

Action 2.3: Create Oasis Innovation Hubs

Objective: Establish innovation hubs in MAB reserves to develop, test, and scale appropriate technologies and management approaches for sustainable oasis development.

Implementation Steps:

1. Select five MAB reserves representing different oasis types to serve as innovation hubs.
2. Develop innovation programs focusing on priority areas such as water efficiency, renewable energy, value-added processing, and ecotourism.
3. Establish partnerships with research institutions, private sector, and local communities to support innovation processes.
4. Implement monitoring and evaluation systems to track innovation outcomes and facilitate learning across hubs.

Timeline: Site selection completed by Q2 2026; innovation programs launched by Q4 2026; first innovation cycle completed by Q4 2027.

Responsible Actors: MAB National Committees, research institutions, private sector partners, local communities.

7.3 Theme 3: Socio-Economic Viability & Livelihoods

Action 3.1: Develop Heritage-Based Value Chains

Objective: Create sustainable value chains for oasis products that recognize and reward cultural heritage values and sustainable production practices.

Implementation Steps:

1. Identify priority products with unique heritage values and market potential, such as specific date palm varieties, medicinal plants, and handicrafts.
2. Develop quality standards and certification systems that recognize traditional production methods and cultural heritage values.
3. Establish partnerships with private sector for product development, marketing, and distribution.
4. Create business incubation services for oasis entrepreneurs, focusing on value-added processing and market access.

Timeline: Product identification completed by Q3 2026; certification systems developed by Q1 2027; business incubation services launched by Q3 2027.

Responsible Actors: Agriculture ministries, private sector partners, business development agencies, local communities.

Action 3.2: Implement Sustainable Tourism Development Program

Objective: Develop sustainable tourism models that showcase oasis cultural and natural heritage while providing direct economic benefits to local communities.

Implementation Steps:

1. Develop sustainable tourism guidelines for oasis systems, focusing on low-impact approaches that respect cultural traditions and environmental limits.
2. Design and pilot tourism products in five MAB reserves, including cultural experiences, agricultural tourism, and nature-based activities.
3. Develop capacity building program for local communities on tourism services, hospitality, and enterprise management.
4. Establish marketing and distribution channels for oasis tourism products, including partnerships with tour operators and online platforms.

Timeline: Guidelines developed by Q4 2026; pilot products launched by Q2 2027; capacity building program implemented by Q4 2027.

Responsible Actors: Tourism ministries, MAB National Committees, tourism operators, local communities.

Action 3.3: Establish Youth Engagement Program

Objective: Create opportunities for youth engagement in oasis conservation and sustainable development, addressing intergenerational knowledge transmission and creating viable livelihoods.

Implementation Steps:

1. Develop youth engagement strategy focusing on education, skills development, entrepreneurship, and cultural revitalization.
2. Establish youth training centers in MAB reserves, offering programs in sustainable agriculture, water management, ecotourism, and value-added processing.
3. Create mentorship programs connecting youth with elders and experts to facilitate knowledge transmission and skill development.
4. Establish seed funding mechanism for youth-led enterprises focused on sustainable oasis development.

Timeline: Strategy developed by Q3 2026; training centers established by Q1 2027; mentorship programs launched by Q2 2027.

Responsible Actors: Education ministries, youth organizations, MAB National Committees, local communities.

This action plan provides a concrete roadmap for implementing the strategic framework, with specific actions, timelines, and responsible actors identified. The plan is designed to be implemented in phases, with initial focus on establishing institutional foundations and knowledge systems, followed by scaling up of interventions based on lessons learned. Regular monitoring and evaluation will be critical to ensure adaptive management and continuous improvement.

Chapter 8

A Framework for Monitoring & Evaluation (M&E)

Effective management of oasis ecosystems requires robust monitoring and evaluation systems that track progress toward sustainability goals and enable adaptive management. The following framework provides specific, measurable indicators across hydrological, ecological, and socio-economic dimensions, with clear data sources to support evidence-based decision-making.

8.1 M&E Framework Structure

The M&E framework is structured around four dimensions that reflect the key aspects of oasis sustainability: hydrological, ecological, socio-economic, and governance. For each dimension, specific Key Performance Indicators (KPIs) have been identified, along with recommended data sources and measurement frequencies.

Figure 8.1: Monitoring and Evaluation Framework for Oasis Sustainability

8.2 Implementation of the M&E Framework

Institutional Arrangements: Effective implementation of the M&E framework requires clear institutional arrangements at national and local levels. At the national level,

a coordinating body should be established with responsibility for overall M&E system design, data aggregation, and reporting. This body should include representatives from water, environment, agriculture, and statistical agencies, as well as research institutions and civil society organizations.

At the local level, M&E units should be established in key oasis systems, with responsibility for data collection, analysis, and reporting. These units should include both technical staff and community representatives, ensuring that local knowledge and perspectives are incorporated into the M&E process.

Data Collection and Management: Data collection should combine conventional scientific monitoring with community-based approaches, creating a comprehensive understanding of system dynamics. Scientific monitoring should include hydrological measurements, ecological surveys, and socio-economic assessments, while community-based monitoring should focus on indicators that can be effectively observed and reported by local residents, such as changes in vegetation patterns, water quality, and the status of cultural practices.

Data management systems should be designed to ensure quality, consistency, and accessibility. This includes standardized protocols for data collection, quality assurance procedures, and secure storage systems. The regional Oasis Knowledge Platform (Action 2.1) should include a dedicated M&E module to facilitate data sharing, analysis, and reporting.

Analysis and Reporting: Data analysis should focus on trends, thresholds, and relationships between different indicators. This includes time-series analysis to track changes over time, spatial analysis to identify patterns and hotspots, and correlation analysis to understand relationships between hydrological, ecological, socio-economic, and governance factors.

Reporting should be tailored to different audiences and purposes, including:

- **Technical Reports:** Detailed analysis for researchers and technical specialists
- **Management Briefs:** Summaries for decision-makers focusing on key findings and recommendations
- **Community Bulletins:** Accessible summaries for local communities highlighting relevant findings and opportunities for engagement
- **Policy Briefs:** Analysis and recommendations for policymakers at national and regional levels

Adaptive Management: The M&E framework should be linked to adaptive management processes that use findings to adjust policies, programs, and interventions. This includes regular review meetings at national and local levels to assess progress, identify challenges, and adjust strategies based on evidence. The framework should also include mechanisms for feedback and learning across different oasis systems, facilitating the exchange of experiences and best practices.

Capacity Building: Effective implementation of the M&E framework requires significant capacity building at national and local levels. This includes training in data collection methods, analysis techniques, reporting skills, and adaptive management approaches. Ca-

pacity building should target both technical staff and community representatives, ensuring that all stakeholders can effectively participate in the M&E process.

The M&E framework should be seen as an integral component of oasis management, rather than an external evaluation process. By providing timely, relevant information on system dynamics and intervention outcomes, it enables evidence-based decision-making and continuous improvement in management approaches.

Chapter 9

Learning from Field Missions: Fayoum, Siwa, and New Valley

The strategic framework developed for oasis conservation is grounded not only in theoretical analysis but also in direct engagement with oasis communities and ecosystems through field missions [6, 4, 5]. These missions to Fayoum University, Siwa Oasis, and the New Valley Governorate provided critical insights into the challenges, opportunities, and practical requirements for implementing sustainable oasis management in Egypt.

Figure 9.1: Field Mission to Fayoum University: Academic Partnership for Oasis Research

9.1 The Fayoum Mission: Building Academic Capacity

The visit to Fayoum University's Faculty of Agriculture revealed the critical importance of academic institutions in oasis conservation. The university has developed significant expertise in date palm genetics, traditional agricultural systems, and water management. However, the mission also highlighted several capacity gaps:

Research Infrastructure: While the faculty possesses strong theoretical knowledge, there are limitations in field research infrastructure and modern analytical equipment. Investment in remote sensing capabilities, soil and water quality laboratories, and long-term monitoring systems would significantly enhance research capacity.

Knowledge Transfer Mechanisms: The gap between academic research and farmer practice remains substantial. Extension services that could translate research findings into practical recommendations for farmers are underdeveloped. This disconnect means that valuable research on improved cultivation techniques, water-efficient irrigation methods, and sustainable pest management often fails to reach the communities that could benefit most.

Interdisciplinary Integration: Oasis management requires integration of hydrology, agriculture, ecology, and social sciences, yet academic programs remain largely siloed. The mission identified the need for interdisciplinary research programs that can address the complex, interconnected challenges of oasis systems.

Figure 9.2: Siwa and New Valley Oases: Field Assessment and Community Engagement

9.2 The Siwa Mission: Traditional Knowledge in Crisis

The visit to Siwa Oasis demonstrated both the resilience and fragility of traditional management systems. Siwa's unique Amazigh culture has maintained sophisticated knowledge systems for millennia, but these are now under severe pressure.

Language and Knowledge Erosion: The Siwi language, which contains precise terminology for traditional water management and agricultural practices, is not being passed on to younger generations. As Arabic becomes dominant and youth migrate to urban areas for education and employment, critical knowledge is being lost. The mission documented numerous examples where specific agricultural techniques, varieties of date palms, or water allocation practices could no longer be explained because the elders who held this knowledge had passed away.

Waterlogging and Salinization: Despite technical interventions in the 1990s that temporarily addressed waterlogging, the problem has re-emerged. The mission found that this was not primarily a technical failure but a governance failure. Without sustained monitoring, enforcement of regulations, and community engagement in management, the physical infrastructure deteriorated and illegal drilling resumed.

Economic Pressures: Traditional agriculture in Siwa is economically marginal. Date palm cultivation, which was once the basis of Siwa's prosperity, now provides minimal income compared to tourism or wage labor. This economic reality drives youth away from agriculture and undermines the transmission of traditional knowledge.

9.3 The New Valley Mission: Scale and Sustainability

The mission to the New Valley Governorate, encompassing the Kharga, Dakhla, and Farafra oases, revealed the challenges of managing oasis systems at scale with non-renewable groundwater resources.

Aquifer Depletion Trajectories: Detailed hydrogeological assessments show that the Nubian Sandstone Aquifer is being extracted at rates far exceeding any potential recharge. Current extraction rates, if continued, will lead to significant declines in water availability within decades. The mission documented cases where deep wells drilled in the 1960s that once flowed artesian now require pumping from depths exceeding 100 meters.

Migration and Development Tensions: Government policies encouraging migration from the overcrowded Nile Valley to the New Valley have created complex tensions. New settlers often lack the traditional knowledge of sustainable oasis agriculture, leading to unsustainable practices. At the same time, these new populations have legitimate development aspirations that must be balanced against sustainability constraints.

Infrastructure and Services: The mission identified significant infrastructure gaps, including inadequate wastewater treatment, limited agricultural extension services, and poor market access for agricultural products. These deficits make it difficult for oasis communities to develop sustainable livelihoods, pushing them toward resource-intensive strategies as the only available path to economic survival.

9.4 Cross-Cutting Lessons from the Missions

Several critical lessons emerged consistently across all three field missions:

1. **Community governance is non-negotiable:** Technical interventions fail without sustained community engagement and governance structures. The Siwa experience demonstrates that even well-designed technical solutions deteriorate without ongoing monitoring, enforcement, and community ownership.

2. **Economics drives behavior:** Sustainability strategies must address economic viability. If traditional practices are economically marginal, they will be abandoned regardless of their ecological value. Creating economic incentives for sustainable practices—through premium markets for traditional products, ecotourism, or payment for ecosystem services—is essential.

Figure 9.3: Geography of The New Valley

3. **Knowledge documentation is urgent:** The loss of traditional knowledge is accelerating. Elders with critical knowledge are passing away, and youth are not learning traditional practices. Systematic documentation must be undertaken immediately, but documentation alone is insufficient—the knowledge must be valued, transmitted, and actively used.

4. **Infrastructure supports sustainability:** Basic infrastructure—wastewater treatment, market access, extension services—is not a luxury but a necessity for sustainable oasis management. Without these foundations, communities have limited options for sustainable development.

5. **Scale matters:** Different oasis systems require different approaches. Small traditional oases like Siwa require strategies centered on knowledge preservation and community governance. Large, modernizing systems like the New Valley require comprehensive planning, infrastructure investment, and managed transitions.

6. **Gender equality is not a social add-on but central to knowledge transmission and system resilience:** Women hold critical traditional knowledge, particularly regarding food processing, seed selection, and household water management. Their systematic exclusion from decision-making processes represents not just a social injustice but a significant knowledge loss.

The lessons learned from Fayoum, Siwa, and the New Valley provide the foundation for scaling oasis conservation efforts across the Arab region. By documenting what works, understanding why it works, and creating frameworks for replication, these missions have established the pathway from strategic vision to operational reality.

The next phase involves establishing the regional expert committee, developing standardized methodologies for documentation and assessment, and creating the financial mechanisms to support implementation. But the fundamental lesson remains: oasis conservation succeeds when it centers community knowledge, creates economic opportunities, and recognizes that heritage is not about preserving the past but about securing the future.

Conclusion

The oases of the Arab region stand at a critical juncture, facing unprecedented challenges from climate change, groundwater depletion, and socio-economic transformations [9]. Yet these systems also represent invaluable repositories of knowledge, culture, and biodiversity that have sustained human communities in arid environments for millennia. This strategic framework provides a comprehensive approach to securing the future of these unique socio-ecological systems, grounded in a paradigm shift that recognizes oases not merely as natural systems but as heritage-based solutions where cultural practices, traditional knowledge, and social structures are integral to ecological resilience.

The framework establishes that the sustainability of oases depends on maintaining their biocultural diversity—the intricate co-evolution between biological systems and human cultural practices. Traditional knowledge systems, sophisticated water management techniques, and social institutions that have sustained these systems for centuries represent critical infrastructure assets that must be preserved, revitalized, and adapted to contemporary challenges. The loss of this knowledge is as direct a threat to oasis sustainability as the depletion of aquifers.

The four pillars of the strategic framework—integrated water management, foundational inventory and typology, centrality of indigenous knowledge, and appropriate use of science and technology—provide a comprehensive approach that addresses the complex interplay of hydrological, ecological, socio-economic, and governance dimensions. The action plan translates this framework into concrete interventions, with specific attention to governance structures, knowledge management systems, and economic incentives that make traditional practices viable in contemporary contexts.

The monitoring and evaluation framework provides the tools for adaptive management, enabling continuous learning and adjustment based on evidence and experience. Together, these elements create a comprehensive approach that can enhance the resilience of oasis systems in the face of mounting challenges.

Preserving oases is not about saving relics of the past but about securing proven models of resilience for the future. As the world increasingly faces challenges of water scarcity, climate change, and sustainable development, the knowledge and practices embedded in oasis systems offer invaluable lessons for adaptation and sustainability. By investing in the unique heritage of oases, the Arab region can preserve not only critical ecosystems but also a proven model of resilience that can guide sustainable development in an increasingly arid world.

The time for action is now. With the establishment of the regional expert committee, the development of comprehensive inventories, and the integration of oases into national and

regional policies [5], we can ensure that these unique systems continue to sustain human communities and biodiversity for generations to come. The challenge is significant, but the strategic framework outlined in this report demonstrates that the solutions are within reach—if we have the wisdom to learn from the past and the courage to embrace a new paradigm for the future.

Appendix A: List of Expert Group Participants

This strategic framework was developed based on the collective expertise of regional and international specialists who participated in the Expert Group Meeting on Managing Groundwater and Dependent Ecosystems in Drylands (Oases), held at the UNESCO Office in Cairo on 23-24 June 2025 [2]. The participants included:

- Dr. Nuria Sanz, Director, UNESCO Regional Office for Egypt and Sudan
- Dr. Mahmoud Fathalla, Director, Directorate of Housing, Water Resources and DRR, League of Arab States
- Dr. Sameh Sakr, UNESCO Cairo Consultant
- Dr. Bisher Imam, UNESCO Regional Hydrologist for the Arab States
- Dr. Abdel Aziz Zaki, UNESCO
- Dr. Pietro Laureano, International Expert, Italy
- Dr. Youssef Brouziyne, IWMI Regional Representative for the Middle East and North Africa (MENA) region
- Dr. Ahmad Hegazy, Expert on Artificial Groundwater Recharge
- Dr. Abdullah Al-Ghafri, Professor of the UNESCO Chair on Aflaj Studies, U.Nizwa, Oman
- Ms. Hanae Lahlou, Water Resources Department, Ministry of Equipment and Water, Morocco
- Dr. Mufleh Alalaween, Expert on Azraq Oasis, Jordan
- Dr. Megbel Aleidan, Expert on AlUla Oasis, Saudi Arabia
- Mohamed Gomaa, Head of Groundwater Central Directorate in the New Valley, Ministry of Water Resources and Irrigation of Egypt
- Dr. Manal Mohamed, Research Institute for Groundwater (RIGW), Egypt
- Representatives from the Egyptian Ministry of Water Resources and Irrigation, and Ministry of Environment
- Experts and representatives from the Arab Water Council, UNESCO IHP National Committee, National Water Research Center, and other regional and academic institutions

Appendix B: Full Comparative Case Study Matrix

This appendix contains the expanded version of the comparative case study matrix presented in Chapter 4, with additional details on each oasis case study, including specific data points, extended analysis of interventions, and more comprehensive lessons learned.

Appendix C: Bibliography of Key Sources and Expert Presentations

This strategic framework synthesizes knowledge from a wide range of sources, including scientific literature, technical reports, and expert presentations [10, 11]. The following key sources provided foundational information and insights:

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[2] UNESCO Regional Office for Egypt and Sudan, "Harnessing heritage for a resilient future: A strategic framework for oasis ecosystems and mab reserves in the arab region," 2025.

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[5] N. Sanz, "Mission report: Unesco visit to the new valley governorate," November 2025.

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